

IT 320 Course Project
Semester-1, 1447H



Software Product Release
WOD

Prepared by

Section #	NAME	ID
79667	Renad Essa Alowais	444200530
	Noura Waleed Alamro	444200941
	Shooq Waleed Alawdah	444201083
	Reema Ibrahim Alraqibah	444201069
	Ghada Saeed Alqahtani	444200371

Supervised By: Dr. Reem AlGifary

Roles And Responsibilities Sheet:

Phase-1

Student name	Role	Reviewer	Reviewer's Feedback
Renad Essa Alowais	• Introduction • Problem • Solution	Reema Ibrahim Alraqibah	Very clear and detailed
Noura Waleed Alamro	• Product Roadmap • Scrum Team	Renad Essa Alowais	All good
Reema Ibrahim Alraqibah	• Vision	Noura Waleed Alamro	All good
Shooq Waleed Alawdah	• Scope	Ghada Saeed Alqahtani	Good and complete work
Ghada Saeed Alqahtani	• Objectives	Shooq Waleed Alawdah	All good

Phase-2

Student name	Role	Reviewer	Reviewer's Feedback
Renad Essa Alowais	• Interviews • Questionnaire • System Users	Reema Ibrahim Alraqibah	All good
Noura Waleed Alamro	• Use case Diagram • Interviews Appendix	Renad Essa Alowais	All good
Reema Ibrahim Alraqibah	• Interviews Appendix • Product Backlog • Interviews Questions	Noura Waleed Alamro	All good
Shooq Waleed Alawdah	• Background • Literature Review • Requirements Elicitation and Analysis • Domain Analysis	Ghada Saeed Alqahtani	All good
Ghada Saeed Alqahtani	• Interview#4 _ Interviews Appendix- Requirements Elicitation _ Jira	Shooq Waleed Alawdah	All good

Phase-3

Student name	Role	Reviewer	Reviewer's Feedback
Renad Essa Alowais	- System Architecture - Login page - Signup page - Friend wishlist page	Reema Ibrahim Alraqibah	All good
Noura Waleed Alamro	- Data Design - Calendar page - Friend profile page	Ghada Saeed Alqahtani	All good
Reema Ibrahim Alraqibah	- Navigation Diagram - Interface Design - UX Guideline - Profile page	Noura alamro	All good
Shooq Waleed Alawdah	- Class diagram - Explore page	Renad Essa Alowais	All good
Ghada Saeed Alqahtani	- Component Design - index.htm page - group_contribution_payments page - group_contribution page - group_dashboard page	Shooq Waleed ALawdah	All good

Phase-4

Student name	Role	Reviewer	Reviewer's Feedback
Renad Essa Alowais	- Chapter 5 Introduction 5.1 User Story Acceptance Testing - User Profile System - User Experience & Navigation	Reema Ibrahim Alraqibah	All good
Noura Waleed Alamro	- User Acceptance Testing (UAT) - Integration Testing & Bug Fixing - Frontend-Backend Integration - Friend Profile & Relationship System - User Experience & Navigation	Ghada Saeed Alqahtani	All good
Reema Ibrahim Alraqibah	5.4 Discussion - Chapter 6 (Conclusion & Future Work) - Calendar & Event Management System - User Experience & Navigation	Noura alamro	All good
Shooq Waleed Alawdah	- User Acceptance Testing (UAT) - Database Schema & Relationships Implementation - Input Validation & Security Implementation - Group Gifting & Contributions System - User Experience & Navigation	Renad Essa Alowais	All good
Ghada Saeed Alqahtani	- Integration Testing - User Search & Discovery Functionality - User Experience & Navigation	Shooq Waleed ALawdah	All good

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1 Chapter 1: Introduction

Gift-giving has long been a cherished tradition that reflects appreciation, connection, and celebration. Whether for birthdays, graduations, weddings, or other personal milestones, a thoughtful gift can strengthen relationships and create lasting memories. However, despite its emotional value, the process of choosing, organizing, and coordinating gifts often becomes stressful. Challenges such as receiving duplicate gifts, forgetting important occasions, or facing confusion in group contributions make the experience less enjoyable and more complicated than it should be.

Across the world, several digital platforms have emerged to simplify the gifting process, such as wish-list applications and local vendor delivery services. While these platforms offer partial solutions-like browsing curated gifts or sending ready-made items-they fail to provide an integrated social experience. Most systems do not allow users to coordinate with friends, track contributions, or reserve items to avoid duplication, resulting in fragmented and impersonal experiences.

Aligned with Saudi Vision 2030, which emphasizes lifestyle improvement through digital transformation, our team developed WOD-an interactive social gifting platform designed to bring convenience, personalization, and cultural relevance to the heart of gift-giving. WOD enables users to create and manage wishlists, browse and interact with posts, search for friends, reserve gifts to prevent duplication, contribute to group gifts, and manage personal profiles. In addition, WOD includes a smart calendar that reminds users of upcoming occasions, ensuring that no important event is forgotten.

Unlike traditional gift or delivery apps, WOD focuses on social connection rather than commerce. Users can explore friends' wishlists and profiles, engage with posts, and plan group gifts-all within a single organized and collaborative platform. This approach transforms gift-giving from a stressful task into a shared, joyful experience that strengthens relationships through thoughtful planning and coordination.

This document details the end-to-end development of the WOD platform. **Chapter 1** has introduced the core challenges in modern gift-giving and positioned WOD as a socially-driven solution. Subsequent chapters will expand on this foundation: **Chapter 2** provides a domain analysis and literature review, **Chapter 3** outlines the requirements engineering process, **Chapter 4** covers the system's design and development, **Chapter 5** discusses the evaluation strategy, and **Chapter 6** concludes with reflections and future work. By presenting this comprehensive plan, we aim to

demonstrate how WOD transforms a universal social practice into a seamless, connected, and joyful digital experience.

1.1 The Problem

Although gift-giving is meant to strengthen social bonds, many individuals encounter recurring challenges:

- **Duplicate gifts:** multiple people unintentionally purchase the same item.
- **Group gifting difficulties:** coordinating contributions across scattered chats and payments leads to confusion and delays.
- **Forgotten occasions:** many people miss birthdays, graduations, or anniversaries without a reliable reminder system.
- **Limited coordination tools:** no simple way to view or reserve friends' wishlists.
- **Basic management gaps:** users often lack proper account tools to manage preferences and interactions.

These issues make gift-giving less enjoyable and more stressful, creating a need for a solution that combines organization, coordination, and convenience.

1.2 The Solution

Our project, **WOD**, presents an integrated digital platform designed to transform the traditional gift-giving experience into an organized, social, and stress-free process. The system directly addresses the core challenges identified—duplication, forgotten occasions, disorganized coordination, and lack of communication—by offering a unified environment where users can manage every stage of the gifting journey from one place.

Through a personalized **user account**, each member can create and maintain a detailed profile that reflects their preferences and activities within the system. This profile becomes the center of interaction, allowing users to share their **posts** and **wishlists**, as well as view and engage with those of their friends.

The **wishlist** feature allows users to add desired gifts, ensuring that friends can easily see what the recipient truly wants. To prevent the common issue of duplicate gifts, WOD introduces a **reservation system** that enables friends to reserve an item from another user's wishlist—instantly marking it as unavailable to others.

For group celebrations, WOD simplifies coordination through a **group gifting** feature, where friends can collectively contribute to a shared gift. The platform transparently tracks contributions and progress, reducing confusion and improving collaboration. Additionally, WOD includes a **calendar** that helps users stay organized and aware of upcoming occasions such as birthdays or graduations.

Unlike many gifting apps that focus solely on transactions or deliveries, WOD prioritizes **social connection and coordination**. Users can **search for friends**, **browse their profiles**, and **interact through posts**, creating a community-driven environment that blends the emotional value of gift-giving with the convenience of technology.

By combining personalization, organization, and connectivity, WOD not only solves the logistical difficulties of gifting but also restores the joy and meaning behind the act itself.

1.2.1 Benefits

The proposed solution offers several tangible benefits to users:

- **Prevents duplication** through the reservation system, ensuring unique and thoughtful gifts.
- **Simplifies group gifting** by providing a clear contribution and tracking interface.
- **Strengthens relationships** through shared posts, wishlists, and friend connections.
- **Enhances organization** with calendar reminders and a centralized gifting hub.
- **Saves time and effort** by combining all gift-planning tools in one accessible platform.
- **Encourages personalization and engagement**, turning gift-giving into a collaborative and enjoyable social experience.

In essence, **WOD** bridges the gap between intention and action—helping users transform gifting from a stressful task into a seamless and meaningful process.

1.3 Product Vision

For individuals and groups of friends

Who struggle to organize, coordinate, and personalize meaningful gifts for special occasions, WOD is an interactive web-based gifting platform.

That simplifies the entire gift-giving process by combining wishlists, reminders, and social coordination in one place.

Unlike traditional wishlist or delivery applications that focus only on purchasing,

Our product emphasizes connection and organization—turning gift-giving into a collaborative and thoughtful experience.

1.4 Product Roadmap

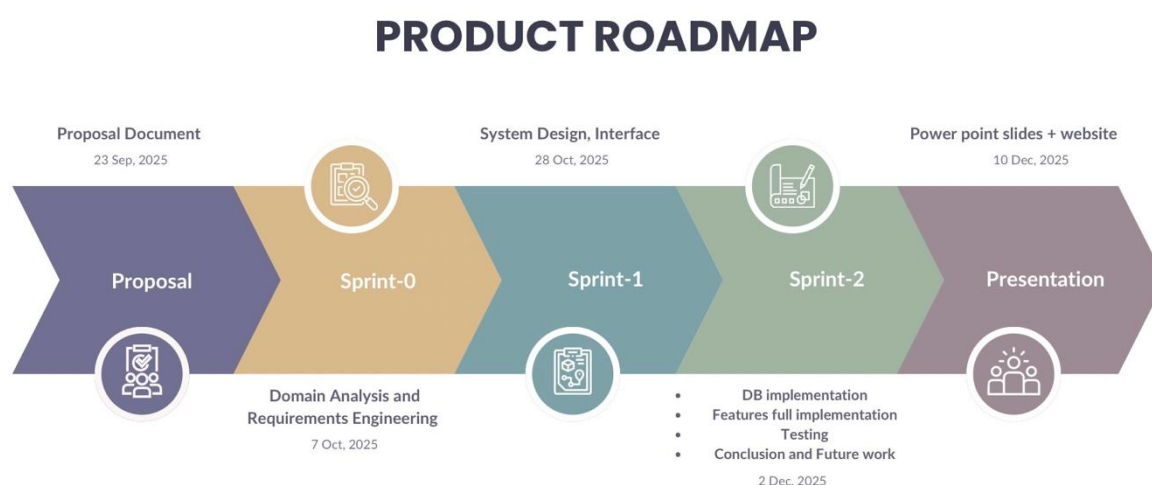


Figure 1: Product Roadmap

1.5 Objectives

This section outlines the product, project, and learning objectives associated with the development of the **“WOD” gifting platform**. These objectives define the goals and intended outcomes to be achieved upon the project’s completion.

Product (customer focus – value):

These objectives describe the core value the product will deliver to users, reflecting how WOD solves real-life challenges in social gift-giving through its main features:

- Allow users to register, log in, and log out securely.
- Enable users to **browse, create, delete, and interact with posts**, fostering a social environment where users can share gift ideas and experiences.
- Support actions such as **liking posts, searching for friends, and adding posts to wishlists** for easier gift discovery.
- Allow users to **view, edit wishlists**, ensuring their preferences remain up-to-date and visible to friends.
- Provide a **reservation system** that lets users reserve gifts from friends’ wishlists to prevent duplication.
- Facilitate **group gifting**, enabling users to contribute collectively to shared gifts in an organized way.
- Allow users to **view and manage their personal profiles**, and explore **friends’ profiles and wishlists** for better coordination.
- Include a **calendar feature** that lets users add and view important occasions, ensuring they never forget special events.

By integrating these features, WOD transforms the gifting process into a connected, collaborative, and enjoyable experience.

Project (solution focus – plan):

These objectives outline the main stages of the project’s development and the steps required to bring the WOD platform to life:

- Conduct domain analysis and review existing social gifting platforms.
- Gather and analyze user requirements through interviews and surveys.
- Define system requirements, architecture, and database design.
- Design user interfaces that support social interaction and usability.
- Implement front-end and back-end functionalities using appropriate development frameworks.
- Integrate key features such as wishlists, posts, reservations, group gifting, and calendars.
- Perform testing to ensure system functionality, performance, and user satisfaction.
- Host and deploy the web-based platform for demonstration and evaluation.

Learning (student focus):

These objectives highlight what our team will learn in order to complete the project, which are:

- Utilizing new tools such as Jira.
- Learn and apply Agile methodology and work within scrum team.
- Learn and use Git and GitHub for version control.
- Learn and apply development frameworks for building a full-stack website.

These objectives guide the project’s direction, ensuring both functionality and team growth. In the next section, we will define the scope of the “WOD” website, outlining its boundaries

1.6 Scope

The WOD platform will be developed as a web-based application designed to run on modern desktop and mobile browsers, using English as the sole language of implementation. The primary users are individuals and small groups seeking to organize and coordinate social gift-giving in a more interactive and structured way.

The project will follow agile development practices and is limited to front-end and back-end web development using commonly available frameworks. Native mobile applications, third-party payment integration, and in-app purchasing are excluded from the initial scope. Notification services will be minimal, with no SMS or push notifications planned. The platform will not include direct integration with external vendors or APIs; all gift purchasing or external interactions will occur outside the system. The focus of this release is to explore core social and technical concepts in web application development within the constraints of available time, tools, and team capabilities.

1.7 The Scrum Team

Scrum Team	
Product Owner:	Renad Alowais
Developers:	Noura Alamro Reema Alraqibah Shooq Alawdah Renad Alowais Ghada Alqahtani
Scrum Master:	Dr. Reem AlGifary
Stakeholder:	Dr. Reem AlGifary

Table 1: The Scrum Team

2 Chapter 2: Domain Analysis

2.1 Background

This section outlines the core concepts and domain context that form the foundation of the WOD platform. It begins with a general overview of the gift-giving domain before detailing the specific solution domain and technological background relevant to our project.

2.1.1 The Gift-Giving Domain

Gift-giving is a universal tradition meant to strengthen social bonds and celebrate special occasions. However, the process is often accompanied by significant challenges that can make it stressful. The core problem domain includes common issues such as receiving duplicate gifts, the difficulty of coordinating group contributions across multiple communication channels, and the frequent occurrence of forgotten important occasions like birthdays and anniversaries. These problems create a clear gap between the intention of giving a meaningful gift and the practical execution.

2.1.2 The Digital Solution Domain

Digital platforms have emerged to address these challenges, primarily through a few key concepts:

- **Wishlist Systems:** A digital wishlist allows users to create and share a list of desired gifts. WOD enhances this concept with a unique gift reservation system to proactively prevent duplication.
- **Group Gifting Coordination:** This feature provides a structured way for multiple people to contribute to a shared gift, moving away from the confusion of scattered chats and payments.
- **Occasion Reminders:** An integrated calendar with reminder functionality helps users stay updated on important dates to ensure no occasion is forgotten.

2.1.3 Technological Background for WOD

Our approach to solving these problems is built on a specific technological foundation:

- **Web-Based Platform:** To ensure wide accessibility, WOD will be developed as a web-based platform. The project scope also includes support for Progressive Web App (PWA) capabilities to offer an app-like experience through a web browser.
- **External Service Integration:** The platform will integrate with local vendor APIs (for flowers, cakes, and chocolates) to allow users to browse gift ideas directly on the platform.
- **Software-Only Solution:** As a purely software-based solution, WOD requires no specialized hardware and is accessible through standard everyday devices like smartphones, tablets, and computers.

This combination of addressing core social challenges with an accessible, web-based technological approach allows WOD to bridge the gap between thoughtful intent and practical execution. To position our solution within the current market landscape, the following section presents a literature review of competing services.

2.2 Literature Review

To position WOD effectively in the market and ensure it addresses the gaps in existing solutions, an analysis of key competitors was conducted. This review examines platforms that currently serve the gift-giving and wishlist space.

Competitor 1: Amazon Wish List [1]



Amazon Wish List is a feature integrated within the vast Amazon e-commerce ecosystem, allowing users to create and share lists of desired products exclusively from Amazon's inventory.

Strengths:

1. Unmatched product selection due to direct integration with the world's largest online retailer.
2. A seamless and trusted purchasing and fulfillment process for users.
3. High brand recognition and a large, established user base.

Limitations:

1. It lacks a dedicated gift reservation system, making duplicate gifts a common occurrence.
2. There is no native functionality to coordinate or track financial contributions for group gifts.
3. The platform does not offer an integrated calendar for occasion reminders.

4. Its scope is limited to Amazon's products, excluding personalized gift options.

Competitor 2: Floward [2]



Floward is an online platform and application specializing in the curated delivery of gifts, flowers, and cakes across the Middle East, with a strong presence in Saudi Arabia.

Strengths:

1. Offers a high-quality, curated selection of gifts from local vendors and artisans.
2. Provides a seamless, end-to-end purchasing and premium delivery experience.
3. Possesses strong brand recognition and trust within the target regional market.

Limitations:

1. Does not support the creation of personal, multi-vendor wishlists by users.
2. Lacks any tools for social coordination, such as group gifting or gift reservation.
3. Operates as a direct vendor, so it cannot display or coordinate gifts from friends' desired lists on other platforms.

Feature	Amazon Wish List	Floward	WOD
Personal Wishlists	yes		yes
Gift Reservation System			yes
Group Gifting Coordination			yes
Integrated Occasion Calendar			yes
Direct Vendor Purchasing	yes	yes	

Table 2: Competing websites and their features comparing with WOD.

From our competitive analysis, we observed that WOD offers a unique value proposition that distinguishes it from established platforms like Amazon Wish List and Floward. Unlike its competitors, WOD is built around a social coordination core, integrating a dedicated gift reservation system and group contribution tracking to directly prevent duplication and simplify collective gifting. Furthermore, while all platforms provide access to gifts, only WOD includes an integrated occasion calendar to help users remember important dates.

These insights have helped us refine our platform's direction by emphasizing features that address the social frustrations of gift-giving, ensuring WOD provides a distinct and more thoughtful user experience centered on coordination rather than just commerce.

In the next chapter, we will explore Requirements Engineering, focusing on the process of defining, analyzing, and documenting the essential system requirements for WOD.

3 Chapter 3: Requirements Engineering

3.1 Requirements Elicitation and Analysis

In this section, we outline the key phases of the requirements engineering process for the WOD platform. Initially, we gathered and documented user requirements through multiple elicitation techniques, including user interviews and questionnaires. These methods helped us capture real user needs, frustrations, and expectations regarding the process of giving and receiving gifts. Additionally, we analyzed the characteristics of our target users and identified the main features required to improve their gift-giving experience.

As part of the requirements elicitation process for the WOD system, we explored several approaches, such as analyzing similar gifting platforms, reviewing online feedback from users of existing apps, and consulting potential stakeholders. However, our primary focus was on interviews and questionnaires, as they provided direct insights from users who frequently buy or coordinate gifts.

For the user interviews, we conducted sessions with five participants (four females and one male) aged between 20 and 32. All participants were active in social and family gifting activities, making them an appropriate representation of WOD's target audience. Each participant was asked a set of structured questions designed to uncover the main challenges they face, such as duplication of gifts, difficulty remembering occasions, and issues in group coordination. The interviews provided valuable qualitative data that guided the system's functional requirements and feature priorities.

The full set of interview questions and detailed answers can be found in [Appendix A](#).

In addition, we distributed an online questionnaire to twelve participants between the ages of 18 and 35 to gather a broader perspective on user experiences. The questionnaire consisted of six closed-ended questions focusing on gift-giving behaviors, common difficulties, and desired improvements

in existing apps. The collected data provided measurable insights into user needs and supported the validation of the interview findings.

The questionnaire form and responses are provided in [Appendix B](#).

During the requirements elicitation process, we relied on various information sources, including **stakeholder input**, **existing gift-giving applications**, and **academic references** on user experience and behavior. These sources allowed us to compare current market gaps with our participants' expectations, ensuring that WOD's proposed features such as **wishlists**, **reminder notifications**, **reservation systems**, and **group contribution management** directly address real user frustrations.

The combined analysis of interviews and questionnaires revealed that most users struggle with **time-consuming gift selection**, **forgetting special occasions**, and **the stress of coordinating group gifts**. Moreover, many participants expressed the desire for a **simple and organized digital solution** that helps them manage gifting processes efficiently while maintaining a personal touch.

Next, we will present the detailed findings from both the **interviews** and the **questionnaire**, highlighting how user insights shaped the design and functionality of the WOD platform.

3.1.1 Interviews

- Participants' Information

This sub-part provides insight into the interview participants. We interviewed five individuals aged between 20 and 32, including four females and one male. All participants are Saudi nationals and represent the primary target group of the WOD platform: socially active young adults who frequently engage in gift-giving with family and friends. Their diverse academic and social backgrounds (students, professionals, and recent graduates) ensured that different perspectives and challenges were reflected in the interviews.

Participant Name	Age	Nationality	Gender
Hadeel ibrahim	20	Saudi	female
Joud Abdullah	27	Saudi	female
Reema Adel	21	Saudi	female
Najoud AlQahtani	32	Saudi	female
Yazed Abdullaziz	23	Saudi	male

Table 3: Overview of participants' age, nationality, and gender in the conducted interviews

- Interviews Results And Findings

This section presents a consolidated analysis of the key findings from five interviews conducted with our target users. The results are organized by interview question to highlight common themes, user frustrations, and opportunities for the WOD platform. Detailed transcripts of each interview are provided in Appendix A.

1st Question: Tell me about the last time you bought a gift for a friend or relative. What was that process like?

The responses revealed a clear divide in the gift-buying experience. A significant theme was the **stress and inconvenience** associated with the process. Participants reported issues with app limitations that led to higher costs, anxiety over choosing an inappropriate gift, and the time-consuming nature of comparing prices. In contrast, participants who had prior knowledge of the recipient's preferences described a **positive and satisfying** experience, emphasizing the value of certainty and personalization. This underscores the need for a platform that reduces uncertainty and saves time.

2nd Question: Have you ever purchased a gift only to discover the recipient already received the same item?

While not universal, the occurrence of **gift duplication** was a significant pain point for those who experienced it. These incidents, which happened during important occasions like baby showers and birthdays, led to strong feelings of **frustration and wasted money**. This finding provides a strong justification for WOD's core feature: a **gift reservation system** to prevent such duplications.

3rd Question: How do you currently remember important birthdays and occasions?

The majority of participants reported **relying on imperfect methods**, such as personal memory, social media notifications, or reminders from others. This often leads to forgotten occasions and subsequent embarrassment. One participant systematically used a digital calendar, confirming the effectiveness of this approach. The consensus indicates that a **built-in occasion reminder calendar** is a crucial and needed feature for WOD.

4th Question: What is the most annoying part of group gift coordination?

Group gifting was consistently described as a source of frustration. The primary challenges identified were **financial disagreements, delayed payments, and difficulties in reaching a consensus** on what gift to purchase. The process of collecting money was highlighted as the most problematic step. These findings directly support the development of WOD's **transparent group contribution tracking system**, which would streamline coordination and financial management.

5th Question: Do you prefer a wishlist or a surprise when buying gifts?

Preferences were split between the desire for **certainty** (wishlists) and the value of **personal touch** (surprises). Some participants favored wishlists for their practicality, while others found more meaning in surprises. Notably, one participant suggested a hybrid approach. This indicates that WOD should support **both models**, perhaps by allowing users to create wishlists while also providing features for friends to coordinate surprise gifts.

6th Question: **When you need to buy a specific local gift (like a custom cake or flowers), what are the three most time-consuming steps you go through right now? Would seeing local vendor catalogs directly within an app eliminate those steps for you?**

Participants uniformly identified three key time-consuming steps:

- 1) comparing prices across different vendors
- 2) selecting specific designs or flavors, and
- 3) arranging for delivery.

All interviewees agreed that integrating **local vendor catalogs** directly into an app would significantly streamline this process, save time, and reduce the stress associated with these tasks.

7th Question: **Is there anything else you would like to add about the gift-giving experience that you struggle with?**

The feedback highlighted a desire for features that go beyond basic purchasing. Participants suggested functionalities for **saving and organizing gift ideas**, **tracking past gifts** to avoid repetition, **personalized recommendations**, and a **social space to share gift experiences**. This points to an opportunity for WOD to evolve into a more comprehensive social gifting hub that fosters inspiration and community.

3.1.2 Questionnaire

We utilized a questionnaire to collect information and better understand users' behaviors, challenges, and preferences related to gift-giving. The questionnaire aimed to support the findings from the interviews and to validate the main issues the WOD platform seeks to solve, such as difficulties in finding meaningful gift ideas, remembering important occasions, and coordinating group gifts.

The survey was conducted online using **Google Forms** and shared through **WhatsApp groups and social media** to reach a wide audience. A total of **12 participants** responded to the questionnaire. All respondents were young adults between the ages of **18 and 35**, which represents WOD's primary target segment. The survey consisted of **six closed-ended questions** designed to identify user needs, pain points, and expectations for a modern gifting platform.

- Questionnaire Results

The results provided clear insights into user challenges and expectations, highlighting several key themes relevant to WOD's concept.

Difficulty in finding meaningful gift ideas:

50% of participants strongly agreed that they often struggle to come up with meaningful gift ideas, while 33.3% agreed and 16.7% were neutral. This shows that most users find it challenging to choose gifts that are both thoughtful and suitable, reinforcing WOD's goal of simplifying and personalizing the gift selection process.

Stress in coordinating group gifts:

58.3% of respondents agreed that organizing contributions and payments for a group gift is stressful or confusing, and 16.7% strongly agreed. This confirms that users need a more

organized and transparent system for handling shared gifts, which supports the inclusion of a **group contribution management** feature in WOD.

Forgetting important occasions:

33.3% of respondents strongly agreed that they sometimes forget important occasions, while 25% agreed and 16.7% somewhat agreed. This finding supports the necessity of adding **automated reminders** and **occasion-tracking tools** to help users stay prepared for birthdays and celebrations.

Difficulty in knowing what others want:

75% of respondents strongly agreed that they wish there was an easier way to know what their friends actually want as gifts, while 25% agreed. This result highlights a significant need for **wishlist-based gifting features**, enabling recipients to express their preferences clearly while still leaving room for surprises.

Challenges in group coordination:

When asked about the biggest frustration in coordinating group gifts, **45.5%** selected *difficulty collecting contributions* and another **45.5%** chose *delays in payments by group members*. In addition, **36.4%** pointed out *disagreements about gift choices*. These findings further support the need for WOD's **digital coordination tools**, which can automate collection and track group progress.

Preferred new features for gifting platforms:

When asked what feature they would like to add to improve gifting apps, **58.3%** selected *a personalized wishlist system*, **50%** preferred *integration with local stores for easy purchases*, **41.7%** wanted *a group contribution system*, and **33.3%** wanted *reminder notifications for important occasions*. This indicates strong demand for a **comprehensive, all-in-one gifting platform** like WOD that combines convenience, personalization, and collaboration.

3.1.3 Summary of Elicitation Finding

The combined data from interviews and the questionnaire reveals a clear and consistent narrative: the current gift-giving process is fraught with uncertainty and inefficiency. Users are deeply frustrated by the guesswork involved in selecting gifts, leading to a strong demand for **wishlist functionality** to provide certainty, while also expressing a desire to preserve elements of personalization and surprise. The coordination of group gifts emerged as a universal pain point, characterized by financial disorganization and decision-making difficulties, underscoring the necessity for a **digital system to manage contributions**. Furthermore, the widespread issue of forgotten occasions validates the need for an **integrated calendar and reminder system**. The significant emotional and financial impact of duplicate gifts also highlights the critical importance of a **reservation feature**. Ultimately, the findings confirm that users are seeking a unified solution to these fragmented challenges, strongly supporting WOD's vision as a comprehensive platform that consolidates wishlists, vendor access, group coordination, and reminders into a single, user-centric hub.

3.2 System Users

This section describes the primary users of the WOD platform. Based on our requirements elicitation, we have identified our target user base and their key characteristics. Understanding these users helps ensure the platform meets their specific needs and capabilities.

The typical WOD user is a socially active young adult who regularly participates in gift-giving occasions with friends and family. These users value meaningful connections but often find the process of selecting and coordinating gifts to be stressful and time-consuming.

Key User Characteristics:

- **Age Range:** 18-35 years old
- **Nationality:** Primarily Saudi Arabian and other Arabic-speaking nationalities
- **Educational Level:** Secondary education or higher
- **Technical Proficiency:** Comfortable with web applications. Their familiarity with social media platforms indicates they are accustomed to core interactive features that WOD will use, such as **managing a personal profile, searching for friends, and navigating a feed of content** (in WOD's case, wishlists and gift ideas).
- **Social Behavior:** Actively maintains personal relationships through social gatherings and gift-giving traditions. They are often part of group chats where gift coordination is currently discussed.
- **Primary Needs:** Seeks to reduce gift-giving stress, save time, prevent duplicate gifts, and ensure their gifts are meaningful and appreciated.

Our user research revealed that these individuals often struggle with remembering important dates, coordinating group gifts across multiple chat applications, and selecting appropriate gifts that recipients will truly appreciate. They are motivated to use digital solutions that can simplify these processes while maintaining the personal touch that makes gift-giving special.

The platform is designed to be intuitive for users with varying levels of technical expertise, ensuring accessibility for both tech-savvy young adults and those less familiar with digital platforms. The interface prioritizes simplicity and clear navigation to accommodate users who may be using such a gift coordination platform for the first time.

In the following section, we will visualize how these users interact with the system through a use case diagram that maps out the core functionalities available to different types of users

3.3 Use Case Diagram

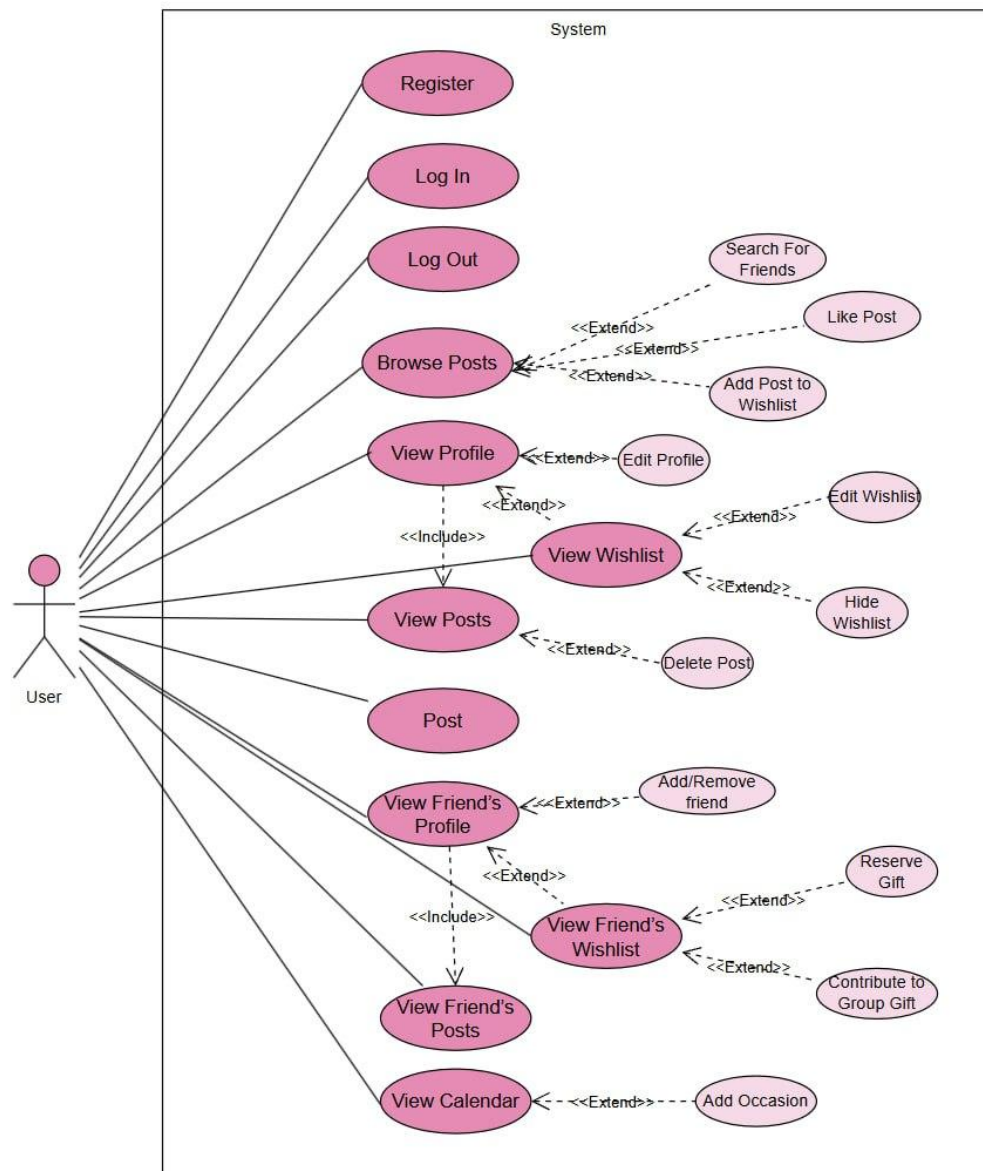


Figure 2: Use Case Diagram of WOD

3.4 Product Backlog

The product backlog for WOD was created and managed using Jira to organize and prioritize all functional and non-functional requirements in the form of user stories. The backlog is structured with the highest-priority, most foundational items at the top, ensuring the development team can begin implementation immediately. The table below represents a snapshot of the prioritized backlog.

PBI	Size	Type	Acceptance Criteria
As a non-registered user, I want to register using my username, email,	4	Feature	As a non-registered user, if I enter a unique username, a valid email, and a

and password so that I can create an account.			password that meets all requirements (min 8 chars, 1 uppercase, 1 number), I should be successfully registered and logged into my account. As a non-registered user, if I enter a username or email that is already taken, I should see an error message: "This username or email is already in use. Please log in or choose another one. As a non-registered user, if I leave any required field empty, I should see an error message: "Please fill in all required fields.
As a registered user, I want to log in using my username or email and password so that the system can verify my identity.	3	Feature	As a registered user, if I enter a correct username/email and password, I should be successfully logged into my account and redirected to the home page. As a registered user, if I enter an incorrect username or password, I should see an error message: "Incorrect username or password. Please try again.
As a registered user, I want to log out of my account to prevent unauthorized access to personal information	2	Feature	As a registered user, when I click the "Log Out" button, I should be logged out and redirected to the login page. Access to personal/account pages should no longer be available unless the user logs in again.
As a user, I want to fully manage my personal Wishlist (create, add, view, edit, and delete items) so that I can keep my preferred gifts up-to-date and share them with friends.	4	Feature	As a user, I should be able to create a new list and add gift items with a title and optional link/image. As a user, I should be able to edit and delete items from my list. As a user, my list should display on my profile and be visible to my approved friends
As a user, I want to control the visibility of my Wishlist so that I can decide whether to use my account for	3	Feature	As a registered user, I should find a "Hide Wishlist" toggle/option in my Profile Settings.

personal gifts or for marketing my content.			As a user, when I activate the " Hide Wishlist " setting, my Wishlist should be invisible to all other viewing users.
As a friend, I want to reserve a gift on a friend's list so that I can prevent gift duplication	4	Feature	As a friend, when I view a friend's list, I should see a "Reserve Gift" button next to available items. As a friend, when I click "Reserve Gift," the item's status should change to "Reserved" and display only to other users who view the list. As a friend, I should be able to unreserve the gift, returning its status to "Available."
As a user, I want to easily contribute to a group gift so that coordinating payments is simple and transparent.	5	Feature	As a user, I should be able to mark a gift item for group contribution. As a user, I should be able to view the total cost, the amount collected so far, and the remaining amount (displayed only to the group members). As a user, I should be able to view a list of friends who have contributed and their amounts. (Note: Payment processing is
As a user, I want to add occasion dates (e.g., birthdays, anniversaries).	3	Feature	As a user, I should be able to add a new occasion (title and date) to my calendar. The calendar view should be readily accessible from the user's profile.
As a user, I want to search for friends by username so that I can view their public Wishlist.	3	Feature	As a user, I should see a prominent search bar dedicated to finding friends. As a user, when I enter a friend's exact username, the system should display their profile. As a user, I should only be able to view a friend's Wishlist if they have set their profile to public or added me as a friend.
As a user, I want a direct button to add a post from the Explore feed to my personal Wishlist.	4	Feature	As a user, I should see a clearly labeled " Add to Wishlist " button (or icon) on every post in the Explore feed.

			As a user, when I click this button, the post details (image, title, link) must be copied immediately to my Wishlist.
As a user, I want to Like an idea on the Explore feed.	4	Feature	As a user, I should see a "Like" button (e.g., a heart icon) on every post in the Explore feed. The number of Likes on the post must be visible, but my individual "Like" action must NOT be publicly visible to my friends or the post creator.
As a Store, I want to include a link to my external shop/contact page so that interested users can easily purchase my products.	2	Feature	As a "Store" user, I should be able to add and update a field for my external shop URL in my profile settings.
As a user or store, I want to post photos and descriptions of gift ideas so that I can share my content on the main Explore feed.	4	Feature	As a registered user/store, I should see a prominent "Create Post" option. The posting interface must allow uploading a photo and adding a title/description. For Stores: The posting interface must include an optional field for adding an external website link. For all users: The post must immediately appear in the main
As a user, I want the system to be available at least 95% of the time so that I am not forced to look for alternative applications.	3	Feature	As a user, if I attempt to access the website 10 times, it should be successfully accessible at least 9 times. All core features (login, viewing lists) must function correctly during periods of high load.
As a user, I want the system to meet performance standards so that I can interact with it smoothly without delays.	3	Feature	As a user, if I open the website, it should load within 3 seconds on a standard internet connection.
As a user, I want the system to be highly usable so that all types of users can easily navigate and interact with it.	2	Feature	As a user, when I visit the website for the first time, I should be able to easily understand the navigation and main features without requiring instructions.

Table 4: Product Backlog

A figure of the live Jira board, reflecting the product backlog, is provided in Appendix C.

4 Chapter 4: System Design and Development

4.1 System Architecture

In this section, we describe the system architecture we adopted for our website, **WOD (Wish Our Day)**. The system architecture defines the organization, scalability, and maintainability of the platform by illustrating how its subsystems interact to achieve all intended functionalities.

After analyzing several design models, we selected the **Client–Server Architecture**, as it provides the best structure for separating the presentation layer (client-side) from the business logic and data management (server-side and database). This separation allows for better system performance, easier updates, and more secure data handling. The architecture consists of three main components: the **Client Side**, the **Server Side**, and the **Database**, which work together through continuous communication via HTTP requests and responses.

4.1.1.1 Client-Side Component

The client-side component of **WOD** serves as the interactive and visual interface that connects users to the platform. It was developed using **HTML** and **CSS** to ensure a clean, user-friendly, and visually appealing experience. Through this layer, users can browse posts, view and search gift ideas, log in, register, and manage their profiles. It also enables users to add or remove posts from their wishlists, view notifications and messages, and navigate easily throughout the platform.

In addition, the client captures user inputs from forms such as registration or login and securely sends them to the server for validation and processing. This design ensures smooth user interaction and efficient communication between the user and the back-end system.

4.1.1.2 Server-Side Component

The server-side component forms the core logic of the **WOD** system.

It is implemented using **PHP** and is responsible for processing client requests, validating user inputs, and managing user sessions. This layer performs various system operations, including **user registration, login and logout**, and **CRUD (Create, Read, Update, Delete)** functions for posts, wishlists, and reservations. It also handles additional system functionalities such as managing group gifting, contributions, and notifications. By serving as the bridge between the client and the database, the server ensures that all operations are securely executed, appropriate data is retrieved or stored, and accurate responses are returned to the client interface.

Through this structured processing, the server upholds data consistency, improves performance, and maintains the reliability of the overall system.

4.1.1.3 Database Component

The database represents the foundation of the **WOD** system, functioning as a permanent and secure data repository. It is implemented using **MySQL** and is designed to handle all the platform's data operations in an organized and scalable manner. The database stores and maintains the following key information:

- **Users:** login credentials, profiles, and preferences.

- **Posts:** gift ideas, descriptions, and images.
- **Wishlists:** personalized lists for each user.
- **Reservations:** booked gifts to prevent duplication.
- **Group contributions:** shared payments for group gifts.
- **Notifications and events:** reminders for special occasions.

This layer ensures that all user and system data is stored consistently, can be efficiently retrieved or updated when needed, and remains secure against unauthorized access.

Its structure supports scalability, making it adaptable for future system enhancements such as integration with external APIs or e-commerce platforms.

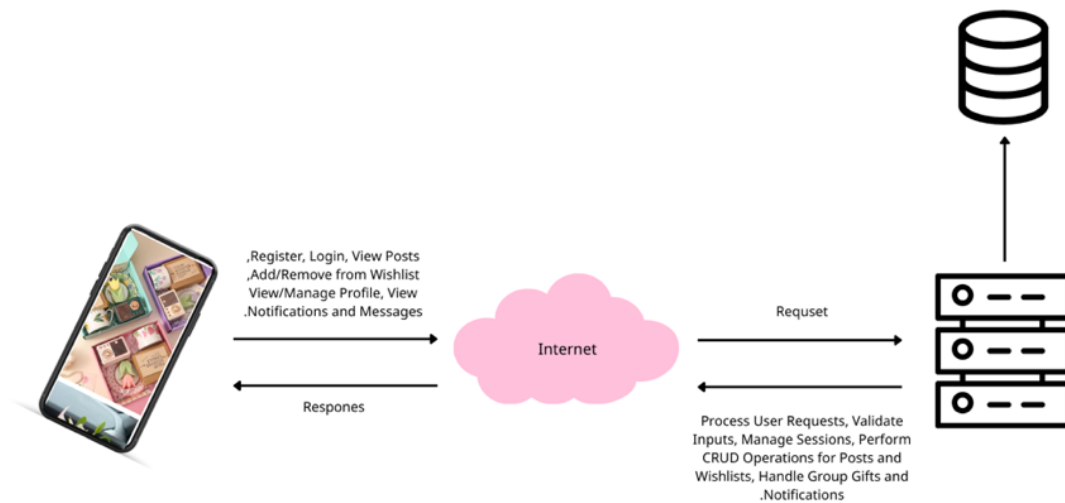


Figure 3: System Architecture

In the next section, we will present the Class Diagram, illustrating the system's architecture and class relationships.

4.2 Class Diagram

This section decomposes the WOD system to understand its structural composition and illustrate how individual components collaborate to deliver core functionality. The class diagram details key classes, their interrelationships, and the methods and attributes that define their behavior.

The system is architected around five cohesive modules that work in concert:

- **User Management & Social Graph:** The User class serves as the central entity, while Friendship manages bidirectional social connections. This social graph enables all subsequent platform interactions.
- **Content & Discovery:** The Post class represents shared gift ideas, and the Like class captures social validation, creating an ecosystem for content discovery and inspiration.
- **Personal Gifting & Coordination:** The WishlistItem class allows users to curate desired gifts, and the Reservation class provides a critical coordination mechanism to prevent duplicate gifting among friends.

- **Event Management:** The CalendarEvent class tracks important occasions, serving as a trigger and reminder for gifting actions.
- **Collaborative Gifting:** The GroupGift, GroupMember, and Payment classes work together to manage collective gift projects, from goal setting and member coordination to financial contributions.

The interrelationships between these classes create a dynamic system where social connections fuel content discovery, which informs personal wishlists. Event management provides context, while collaborative tools enable group gift execution. This decomposition clearly demonstrates how WOD's components work together to provide a holistic social gifting experience.

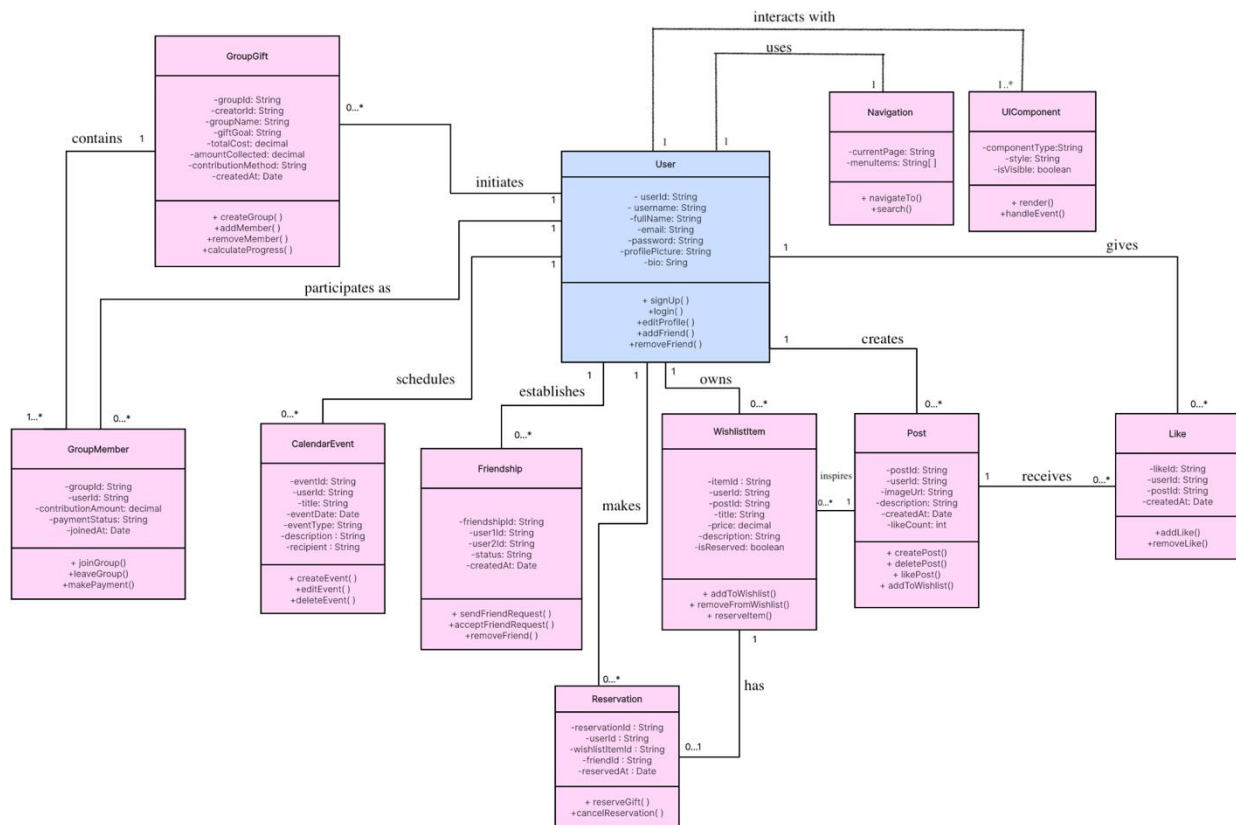


Figure 4: Class Diagram

4.3 Data Design

- EER Diagram

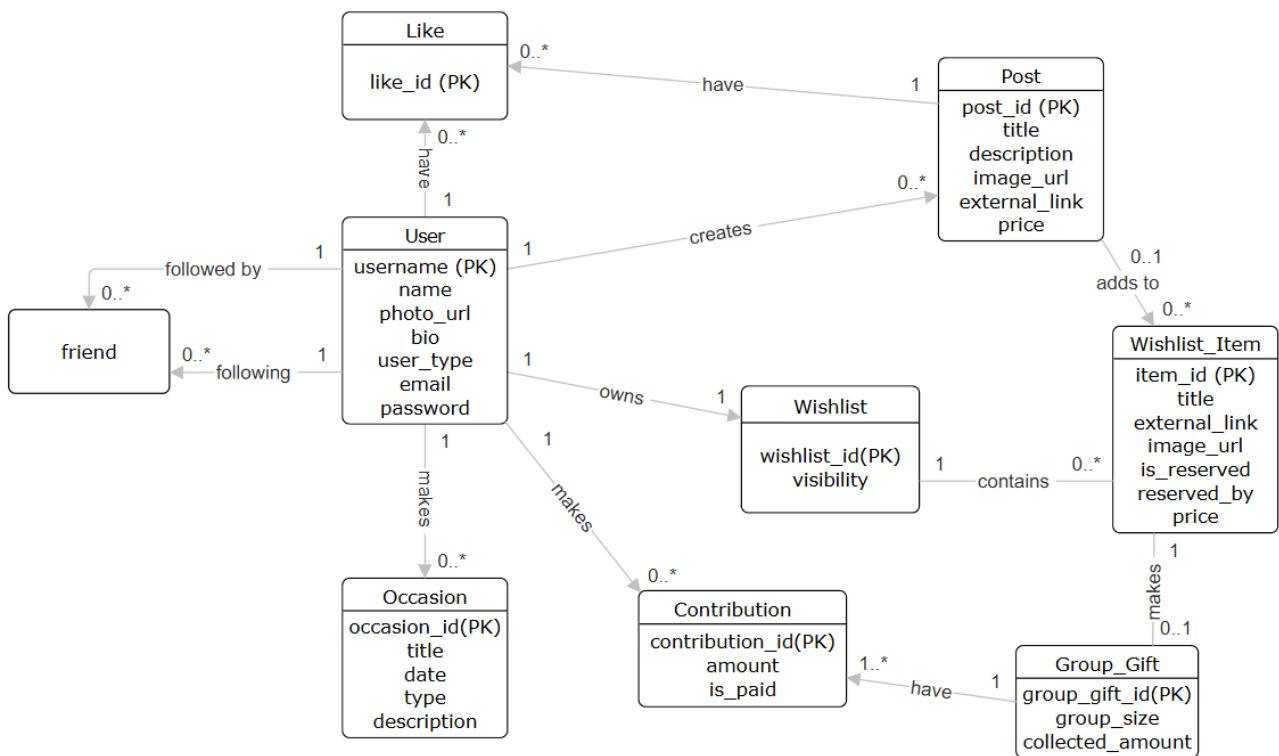


Figure 5: EER Diagram

- Schema

User (username, name, photo_url, bio, user_type, email, password)
Primary Key: username

Post (post_id, title, description, image_url, external_link, price, username)
Primary Key: post_id

Foreign Key: username References User (username)

Like (like_id, username, post_id)

Primary Key: like_id

Foreign Key: username References User (username)

Foreign Key: post_id References Post (post_id)

Friend (username_1, username_2)

Primary Key: (username_1, username_2)

Foreign Key: username_1 References User (username)

Foreign Key: username_2 References User (username)

Occasion (occasion_id, title, date, type, description, username)

Primary Key: occasion_id

Foreign Key: username References User (username)

Wishlist (wishlist_id, visibility, username)

Primary Key: wishlist_id

Foreign Key: username References User (username)

Wishlist_Item (item_id, title, external_link, image_url, is_reserved, price, reserved_by, wishlist_id, post_id)

Primary Key: item_id

Foreign Key: reserved_by References User (username)

Foreign Key: wishlist_id References Wishlist (wishlist_id)

Foreign Key: post_id References Post (post_id)

Group_Gift (group_gift_id, group_size, collected_amount, item_id)

Primary Key: group_gift_id

Foreign Key: item_id References Wishlist_Item (item_id)

Contribution (contribution_id, amount, is_paid, username, group_gift_id)

Primary Key: contribution_id

Foreign Key: username References User (username)

Foreign Key: group_gift_id References Group_Gift (group_gift_id)

- Data Dictionary

Data Dictionary showing description of all entities:

Entity Name	Description	Aliases	Occurrence
-------------	-------------	---------	------------

User	An individual who uses the system, creates posts, has friends, creates occasions, owns wishlist, likes posts, and may contribute to group gifts.	-	A User must have a unique username.
Post	Content created by a user that others can like and which may optionally add an item to a wishlist.	-	A Post is created by one User and can have zero or more Likes.
Like	Represents a user's appreciation for a post.	-	A Like is for one Post and is made by one User.
Friend	Represents a directional "followed by" or "following" relationship between two Users.	-	A User can follow zero or more other Users and be followed by zero or more Users.
Occasion	A specific event or celebration recorded by a user, which may relate to gift-giving.	-	An Occasion is made by one User.
Wishlist	A collection of desired items created and owned by a user, with defined visibility.	-	A Wishlist is owned by one User and contains zero or more Wishlist_Items.
Wishlist_Item	An item listed on a Wishlist. It can be reserved by another user and may become part of a Group_Gift.	-	A Wishlist_Item belongs to one Wishlist and may be added from zero or one Post. It can be part of zero or one Group_Gift.
Group_Gift	A mechanism for multiple users to contribute money to purchase a specific Wishlist_Item.	-	A Group_Gift is associated with one Wishlist_Item and receives one or more Contributions.
Contribution	A financial donation made by a User towards a specific Group_Gift.	-	A Contribution is made by one User toward one Group_Gift.

Table 5:: Data Dictionary showing description of all entities

Data Dictionary showing description of all relationships:

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity	Description
User	1	creates	Post	0..*	Represents a User publishing content.

User	1	owns	Wishlist	1	Represents a User's personal list of desired items.
User	1	makes	Occasion	0..*	Represents a User creating a record of an event.
User	1	have	Like	0..*	Represents a User expressing approval for a Post.
Post	1	have	Like	0..*	Represents a Post receiving approval from Users.
Post	0..*	adds to	Wishlist_Item	0..1	Represents a Post suggesting an item for a Wishlist.
User	0..*	followed by	Friend	0..*	Represents the unidirectional relationship where one User is followed by another.
User	0..*	following	Friend	0..*	Represents the unidirectional relationship where one User is following another.
Wishlist	1	contains	Wishlist_Item	0..*	Represents an Item being part of a specific Wishlist.
User	0..1	reserved by	Wishlist_Item	0..*	Represents a User committing to purchase an Item for the Wishlist owner.
Wishlist_Item	1	makes	Group_Gift	0..1	Represents a Wishlist_Item being funded via a Group_Gift.
Group_Gift	1..*	have	Contribution	0..*	Represents the collection of funds for a Group_Gift.
User	1	makes	Contribution	0..*	Represents a User donating funds to a Group_Gift.

Table 6: of description of all relationships

Data Dictionary showing description of all attributes:

Entity Name	Attribute	Description	Data Type	Length	Nulls	Multi-Valued	Default Value	Range	PK
User	username	Unique identifier and login name for the User.	VARCHAR	-	N	N	-	-	Y
	name	The User's full name.	VARCHAR	-	N	N	-	-	N
	photo_url	URL to the User's profile photo.	VARCHAR	-	Y	N	-	-	N
	bio	Short description or biography of the User.	VARCHAR	-	Y	N	-	-	N
	user_type	Categorization of the User (user, store).	VARCHAR	-	Y	N	'user'	-	N
	email	User's email address.	VARCHAR	-	N	N	-	-	N
	password	User's password.	VARCHAR	-	N	N	-	-	N
Post	post_id	Unique auto-increment ID for the Post.	INTEGER	-	N	N	Auto	-	Y
	title	Title of the post.	VARCHAR	-	N	N	-	-	N
	description	Detailed content of the post.	VARCHAR	-	Y	N	-	-	N
	image_url	URL for the main image in the post.	VARCHAR	-	Y	N	-	-	N
	external_link	A link to an external source (e.g., a product page).	VARCHAR	-	Y	N	-	-	N
	price	The price of the item/product mentioned in the post.	DECIMAL/FLOAT	-	Y	N	-	>=0	N
	username	Foreign key linking to the User who created the Post.	VARCHAR	-	N	N	-	-	N
Like	like_id	Unique auto-increment ID for the Like.	INTEGER	-	N	N	Auto	-	Y
	username	Foreign key linking to the User who liked the Post.	VARCHAR	-	N	N	-	-	N
	post_id	Foreign key linking to the Post that was liked.	INTEGER	-	N	N	-	-	N

Friend	username_1	Foreign key for the User who is <i>following</i> (the primary user).	VARCHAR	-	N	N	-	-	Y
	username_2	Foreign key for the User who is <i>followed</i> (the secondary user).	VARCHAR	-	N	N	-	-	Y
Occasion	occasion_id	Unique auto-increment ID for the Occasion.	INTEGER	-	N	N	Auto	-	Y
	title	Title of the occasion (e.g., "Birthday").	VARCHAR	-	N	N	-	-	N
	date	The date of the occasion.	DATE	-	N	N	-	-	N
	type	Category of the occasion (e.g., "Holiday", "Milestone").	VARCHAR	-	Y	N	-	-	N
	description	Details about the occasion.	VARCHAR	-	Y	N	-	-	N
	username	Foreign key linking to the User who created the Occasion.	VARCHAR	-	N	N	-	-	N
Wishlist	wishlist_id	Unique auto-increment ID for the Wishlist.	INTEGER	-	N	N	Auto	-	Y
	visibility	Controls if others can view the Wishlist.	BOOLEAN	-	N	N	T	T/F	N
	username	Foreign key linking to the User who owns the Wishlist.	VARCHAR	-	N	N	-	-	N
Wishlist_Item	item_id	Unique auto-increment ID for the Wishlist Item.	INTEGER	-	N	N	Auto	-	Y
	title	Name of the desired item.	VARCHAR	-	N	N	-	-	N
	external_link	Direct link to purchase the item.	VARCHAR	-	Y	N	-	-	N
	image_url	URL for the item's image.	VARCHAR	-	Y	N	-	-	N
	is_reserved	Boolean flag indicating if the item has been reserved by a user.	BOOLEAN	-	N	N	F	T/F	N
	price	Cost of the item.	DECIMAL/FLOAT	-	Y	N	-	>=0	N

	reserved_by	Foreign key linking to the User who reserved the item (NULL if not reserved).	VARCHAR	-	Y	N	-	-	N
	wishlist_id	Foreign key linking to the Wishlist this Item belongs to.	INTEGER	-	N	N	-	-	N
	post_id	Foreign key linking to the Post that was liked.	INTEGER	-	N	N	-	-	N
Group_Gift	group_gift_id	Unique auto-increment ID for the Group Gift.	INTEGER	-	N	N	Auto	-	Y
	group_size	The planned number of people needed to contribute.	INTEGER	-	Y	N	-	> 0	N
	collected_amount	The total amount of money contributed so far.	DECIMAL/FLOAT	-	N	N	0.00	>= 0	N
	item_id	Foreign key linking to the Wishlist Item being funded.	INTEGER	-	N	N	-	-	N
Contribution	contribution_id	Unique auto-increment ID for the Contribution.	INTEGER	-	N	N	Auto	-	Y
	amount	The monetary amount contributed by the User.	DECIMAL/FLOAT	-	N	N	-	> 0	N
	is_paid	Boolean flag indicates if the contribution payment has been successfully processed.	BOOLEAN	-	N	N	F	T/F	N
	username	Foreign key linking to the User who made the Contribution.	VARCHAR	-	N	N	-	-	N
	group_gift_id	Foreign key linking to the Group Gift receiving the contribution.	INTEGER	-	N	N	-	-	N

Table 7: description of all attributes

4.4 Component Design

In this section, we will look at our system's component design, focusing on the functionality for reserving gifts, viewing a friend's wishlist, and adding a post to the wishlist. We will give activity

diagrams and pseudocode for each key feature, defining the logic and steps necessary for executing them.

- **Reserve Gift**

Classification: Function

Definition: As a friend, I want to reserve an item on a friend's Wishlist to change its status and prevent duplication by other gift givers

Pseudocode:

FUNCTION Reserve_Gift(GiftID, ReservingUserID)

 DISPLAY Friend's Wishlist with item statuses

 WHILE NOT User_Clicks("Reserve") DO

 CONTINUE

 END WHILE

 LOCK Gift_Record(GiftID)

 IF Get_Gift_Status(GiftID) == "Available" THEN

 Set_Gift_Status(GiftID, "Reserved")

 Set_Reserved_By(GiftID, ReservingUserID)

 UNLOCK Gift_Record(GiftID)

 DISPLAY "Confirmation: Gift has been successfully reserved."

 ELSE

 DISPLAY "Error: This gift is already reserved by someone else."

 UNLOCK Gift_Record(GiftID)

 END IF

END FUNCTION

Activity Diagram:

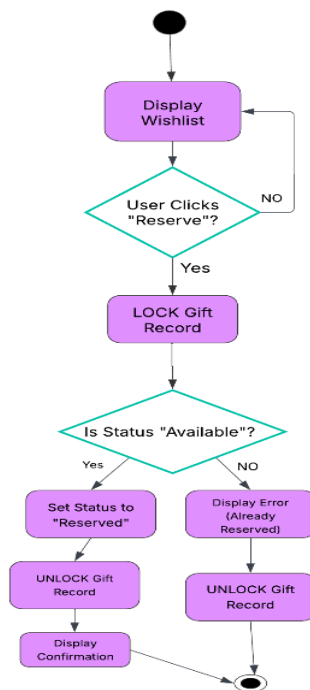


Figure 6: Activity Diagram

- View Friend's Wishlist

Classification: Function

Definition : As a user, I want to search for friends and view their public Wishlist, respecting the visibility settings to ensure privacy and coordination

Pseudocode:

FUNCTION View_Friend_Wishlist(CurrentUserID, SearchUsername)

DISPLAY "Search Bar"

INPUT SearchUsername

FriendUserID = Find_User_By_Username(SearchUsername)

IF FriendUserID IS NULL THEN

DISPLAY "Display Error: User not found"

RETURN

END IF

```
Open_Profile_Page()
Wishlist_Settings = System_Retrieve_Privacy_Settings(FriendUserID)

IF Wishlist_Settings.IsHidden == TRUE THEN
    DISPLAY "Display Error: Wishlist Is Private"
    RETURN
END IF

IF Wishlist_Settings.Visibility == "FriendsOnly" THEN

    IF NOT Is_Current_User_A_Friend(CurrentUserID, FriendUserID) THEN
        DISPLAY "Display Error: Must be Friends"
        RETURN
    END IF

END IF

Display_Wishlist(FriendUserID)

END FUNCTION
```

Activity Diagram:

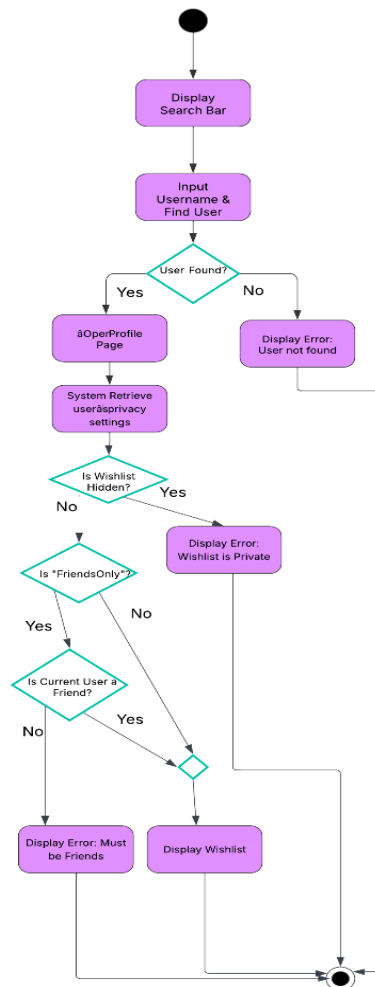


Figure 7: Activity Diagram

- Add Post to Wishlist

Classification: Function

Definition: As a user, I want a direct button to add a post from the Explore feed to my personal Wishlist so that the post details (image, title, link) must be copied immediately to my Wishlist.

Pseudocode:

FUNCTION Add_Post_To_Wishlist(PostID, UserID)

DISPLAY "Explore Feed"

IF User clicks "Add to Wishlist" on PostID THEN

PostDetails = Get_Post_Details(PostID)

```
IF PostDetails IS NULL OR PostDetails.Title IS NULL THEN  
    DISPLAY "Error: Retrieval Failed."  
    RETURN
```

```
END IF
```

```
Create_Wishlist_Item(  
    UserID,  
    PostDetails.Title,  
    PostDetails.Image,  
    PostDetails.ExternalLink,  
    Status: "Available"  
)
```

```
    DISPLAY "Success: Post added to your Wishlist."
```

```
END IF
```

```
RETURN
```

```
END FUNCTION
```

Activity Diagram:

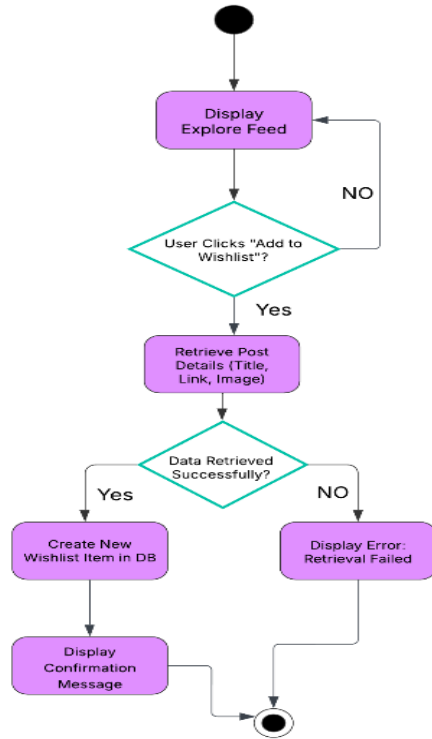


Figure 8:Activity Diagram

4.5 Interface Design

we present the **Interface Design** of our system, focusing on the website structure and user navigation flow. We use a navigation diagram to illustrate how users move between different screens and interact with key **gifting and coordination features**. Additionally, we highlight the UX guidelines that were incorporated into the design to enhance **usability and eliminate logistical friction**. These considerations ensure that the interface is **intuitive, well-structured, and supports seamless user interactions** throughout the collaborative gift-giving process.

The navigation flow starts on the **index page**, where users can log in or create an account. After signing in, they are directed to the **Posts** page. A sidebar provides quick access to the **Home (Posts)** page, the **Calendar** for adding events, the **Profile** to view posts and manage or hide the wishlist, and the **Group Gift Dashboard** to track active contributions and pending invitations. From the Posts page, users can also search for friends, visit their profiles, and view their posts and wishlists.

4.5.1.1 Navigation Diagram

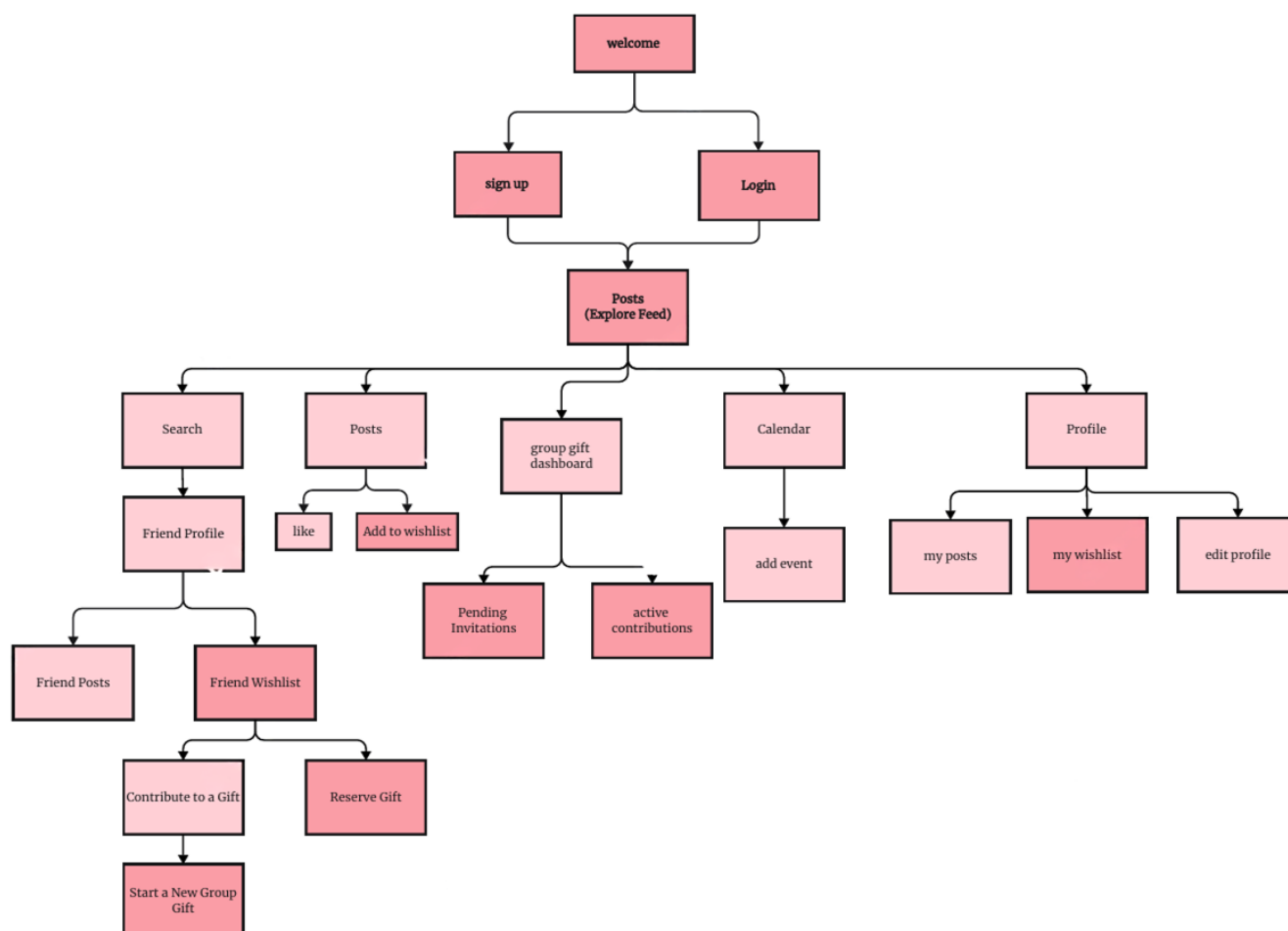


Figure 9:Navigation Diagram

4.5.1.2 UX Guidelines implemented for each interface:

Consistent Design

We maintain a cohesive design across the website to ensure clarity and familiarity. Unified visual elements—such as a consistent color palette, fonts, and layout—create a smooth and aesthetically pleasing browsing experience across all pages.

Clear Navigation

The website includes a simple and intuitive navigation system. A sidebar with clearly labeled sections like Home (Posts), Calendar, Profile, and Group Gift Dashboard helps users move easily between pages without confusion.

Feedback and Confirmation

Users receive instant feedback when interacting with the interface. For example, when creating a new event or managing group gift contributions, the system displays confirmation messages to ensure the action has been completed successfully.

Flexibility and Efficiency

The platform is designed to allow users to perform tasks quickly and efficiently. Users can easily search for friends, view their profiles, check Wishlist, and manage their gifting activities, all within a few clicks.

Error Handling

We prevent errors through input validation and clear prompts. If users attempt an incorrect action, such as submitting incomplete event details, the system notifies them and guides them on how to correct it.

4.5.2 Interface Design Links :

GitHub Link:

<https://github.com/NouraaaW/WOD>

Jira Link:

<https://iirenadii.atlassian.net/jira/software/projects/IT320/boards/3/backlog?atlOrigin=eyJpIjoiZWU5MjNkOWEzYWQ3NDcwOTgyNTcwYWVmNzNmNjlkMjciLCJwIjoiaj9>

4.6 Implementation:

In the implementation phase, we developed WOD as a responsive web application using modern front-end technologies including HTML5, CSS3, and JavaScript. The platform was structured as a Progressive Web App (PWA) to ensure cross-device compatibility and an app-like user experience accessible through standard web browsers without installation requirements.

The front-end architecture follows a component-based approach, with modular CSS styling that maintains visual consistency across all pages including authentication, social feed, wishlist management, and profile sections. We implemented interactive features such as the Pinterest-style grid layout for gift inspiration, hover effects for post engagement, and responsive navigation elements that adapt to various screen sizes.

For data persistence and user management, the implementation focuses on client-side storage and session management, allowing users to interact with core features including wishlist creation, gift reservation, and social interactions. The interface design prioritizes intuitive navigation with a fixed sidebar for quick access to key functionalities and a centralized search system for discovering gifts and friends.

The platform's implementation successfully demonstrates all core features outlined in our requirements, including user registration and authentication, dynamic wishlist management with reservation capabilities, social feed interactions, and basic occasion reminder functionality. This foundation provides a solid base for future backend integration and enhanced feature development while delivering a fully functional user experience that addresses the key pain points in gift-giving coordination identified in our research.

The implementation maintains alignment with Saudi Vision 2030's digital transformation goals by providing an accessible, user-friendly platform that simplifies the traditional gift-giving process through modern web technologies and social coordination features.

5 Chapter 5: System Evaluation

This chapter documents the testing conducted on the final implemented version of the WOD system. The goal of this chapter is to verify that every functional and non-functional requirement has been successfully met and that the system performs as expected from both a technical and user-experience perspective.

The testing activities included User Story Acceptance Testing, Integration Testing, and User Acceptance Testing (UAT). These testing methods ensure that individual features work correctly, components interact smoothly when integrated, and real users from the target audience can use the system effectively.

5.1 User Story Acceptance Testing

User Story Acceptance Testing aims to confirm that each user story fulfills its intended purpose from the user's perspective. It answers the question:

“Does the system meet the acceptance criteria of each user story when executed by the user?”

To perform this testing, we identified all 16 user stories defined earlier in the project. For each story, we extracted its acceptance criteria and designed a detailed test case that simulates realistic user interaction with the WOD system.

All user stories were implemented and tested in **Sprint 2**, as required by Phase 4. Each test case contains step-by-step actions and expected outcomes. A “Pass” status indicates that the implemented feature meets the acceptance criteria and behaves as expected.

The following table documents the acceptance testing results for all WOD user stories.

Sprint Number	User story	Acceptance criteria	Test cases(s)	Pass?	Comments
2	As a non-registered user, I want to register using my username, email, and password so that I can create an account.	1. If I enter a unique username, valid email, and a strong password, I should be successfully registered and logged in. 2. If the username/email is taken, I should see an error. 3. If a required field is empty, I	1. Attempt to register with valid, unique credentials. 2. Attempt to register with an already registered email address. 3. Attempt to register leaving a mandatory field empty. Expectation: The system handles valid input by	Pass	

		should see an error.	successfully registering and handles invalid/incomplete input by displaying appropriate error messages.		
2	As a registered user, I want to log in using my username or email and password so that the system can verify my identity.	1. If I enter correct credentials, I should be successfully logged into my account and redirected to the home page. 2. If I enter incorrect credentials, I should see an error message.	1. If I enter correct credentials, I should be successfully logged into my account and redirected to the home page. 2. If I enter incorrect credentials, I should see an error message.	Pass	
2	As a registered user, I want to log out of my account to prevent unauthorized access to personal information.	• Logout button visible only when logged in. • Clicking logout ends the session. • User is redirected to login page. • Restricted pages become inaccessible.	1. Log in successfully and click the dedicated "Log Out" button. 2. Immediately attempt to navigate to a restricted page (e.g., Profile or Wishlist) using the URL. 3. Expected Outcome: The user MUST be logged out and redirected to the Login page (Criterion 1). The system MUST prevent access to restricted pages in step 2 (Criterion 2).	Pass	
2	As a user, I want to fully manage my	1. I should be able to create a new list and add	1. Create a new Wishlist item with a title, image, and	Pass	

	personal Wishlist (create, add, view, edit, and delete items) so that I can keep my preferred gifts up-to-date and share them with friends.	gift items with a title and optional link/image. 2. I should be able to edit and delete items from my list. 3. My list should display on my profile and be visible to my approved friends.	link. 2. Edit the title of the newly added item. 3. Delete the item. 4. Log in as a friend and view the list. Expectation: All CRUD operations (Create, Read, Update, Delete) are successful, and the final list is visible to approved friends.		
2	As a user, I want to control the visibility of my Wishlist so that I can decide whether to use my account for personal gifts or for marketing my content.	1. I should find a "Hide Wishlist" toggle/option in my Profile Settings. 2. When I activate the "Hide Wishlist" setting, my Wishlist should be invisible to all other viewing users.	1. Locate and activate the "Hide Wishlist" option in profile settings. 2. Log in as a separate user (e.g., a friend) and attempt to access the Wishlist. Expectation: The Wishlist is inaccessible to all other users when the setting is active.	Pass	
2	As a friend, I want to reserve a gift on a friend's list so that I can prevent gift duplication.	1. When I view a friend's list, I should see a "Reserve Gift" button next to available items. 2. When I click "Reserve Gift," the item's status should change to "Reserved" and display only to other users who view the list. 3. I should be able to unreserve the gift	1. As User A, reserve an available item on User B's list. 2. Log in as User C and confirm the item status is "Reserved." 3. As User A, locate the item and click "Unreserve." Expectation: The status changes accurately between "Available" and "Reserved," and the mechanism prevents other	Pass	

			friends from reserving a reserved item.		
2	As a user, I want to easily contribute to a group gift so that coordinating payments is simple and transparent.	1. I should be able to mark a gift item for group contribution. 2. I should be able to view the total cost, the amount collected so far, and the remaining amount (displayed only to the group members). 3. I should be able to view a list of friends who have contributed and their amounts.	. Create a group gift and specify the target cost. 2. Log in as two separate friends and make different contributions. 3. View the group gift dashboard. Expectation: The dashboard correctly sums the contributions, calculates the remaining amount, and lists the contributors transparently.	Pass	
2	As a user, I want to add occasion dates (e.g., birthdays, anniversaries).	1. I should be able to add a new occasion (title and date) to my calendar. 2. The calendar view should be readily accessible from the user's profile.	1. Navigate to the "Calendar" feature. 2. Add a new occasion with a title and future date. 3. Verify the occasion is stored and appears in the calendar view. Expectation: The new occasion is successfully recorded and visible in the accessible calendar interface	Pass	
2	As a user, I want to search for friends by username so that I can view their public Wishlist.	1. I should see a prominent search bar dedicated to finding friends. 2. When I enter a friend's exact username, the	1. Use the search function to look up a friend by username. 2. Navigate to the friend's profile page. 3. Verify that if the	Pass	

		system should display their profile. 3. I should only be able to view a friend's Wishlist if they have set their profile to public or added me as a friend	Wishlist is set to "Friends Only," a non-friend cannot view it. Expectation: Search results are accurate, and Wishlist visibility respects the friend's privacy settings.		
2	As a user, I want a direct button to add a post from the Explore feed to my personal Wishlist.	1. I should see a clearly labeled "Add to Wishlist" button (or icon) on every post in the Explore feed. 2. When I click this button, the post details must be copied immediately to my Wishlist.	1. On the Explore Feed, click the "Add to Wishlist" button on a featured post. 2. Navigate to your Wishlist. Expectation: A new item is automatically created in the Wishlist, populated with the image, title, and link from the original post	Pass	
2	As a user, I want to Like an idea on the Explore feed.	1. I should see a "Like" button (e.g., a heart icon) on every post in the feed. 2. The number of Likes on the post must be visible, but my individual "Like" action must NOT be publicly visible to my friends or the post creator.	1. Click the "Like" button on a post. 2. Verify that the count increases. 3. Confirm that the names/identity of users who liked the post are not displayed publicly. Expectation: The Like count updates accurately, and the user's anonymity is maintained as required.	Pass	
2	As a Store, I want to include a link	1. As a "Store" user, I should be able to add	1. Log in as a Store user and access profile	Pass	

	to my external shop/contact page so that interested users can easily purchase my products.	and update a field for my external shop URL in my profile settings.	settings. 2. Add the external shop URL and save changes. 3. View the Store profile as a regular user. Expectation: The external shop link is correctly displayed and clickable on the Store's profile page.		
2	As a user or store, I want to post photos and descriptions of gift ideas so that I can share my content on the main Explore feed.	1. I should see a prominent "Create Post" option. The posting interface must allow uploading a photo and adding a title/description. 2. For Stores: The posting interface must include an optional field for adding an external website link. 3. The post must immediately appear in the main Explore feed	1. Create and submit a post as a regular user. 2. Create and submit a post as a Store user, including an external link. 3. Check the Explore Feed. Expectation: Both posts appear correctly formatted and immediately visible in the Explore Feed.	Pass	
2	As a user, I want the system to be available at least 95% of the time so that I am not forced to look for alternative applications.	1. If I attempt to access the website 10 times, it should be successfully accessible at least 9 times. 2. All core features (login, viewing lists) must function correctly during	1. Monitor website uptime over a testing period. 2. Simulate high traffic load on core features (Login, Wishlist). Expectation: The system maintains availability and reliability above the 95% threshold	Pass	

		periods of high load.	under varying conditions.		
2	As a user, I want the system to meet performance standards so that I can interact with it smoothly without delays.	1. If I open the website, it should load within 3 seconds on a standard internet connection.	1. Measure the load time of the main Explore Feed page using standard developer tools. Expectation: The page loads and becomes fully interactive in under 3 seconds.	Pass	
2	As a user, I want the system to be highly usable so that all types of users can easily navigate and interact with it.	1. When I visit the website for the first time, I should be able to easily understand the navigation and main features without requiring instructions	1. Conduct an observation test where a first-time user attempts a core task (e.g., reserving a gift). Expectation: The user completes the task successfully with minimal confusion or guidance.	Pass	

Table 8: User Story Acceptance Testing

5.2 Integration Testing

In this section, integration testing was performed to ensure that system components function correctly when combined. The focus was on verifying the interaction between modules after each newly added feature. We adopted a step-by-step, incremental approach, integrating key components such as authentication, content discovery, and core gifting functionalities. For each integration step, specific test cases were designed and executed to validate data flow and ensure consistent system behavior, directly following the defined Integration Hierarchy for WOD.

Integration Hierarchy

5.2.1 In this sub-section, we present the integration hierarchy diagram for WOD (Wish Our Day), which illustrates the structure and sequence in which the system's components were integrated and tested to build the full functionality of the website. The numerical labels on the hierarchy diagram indicate the order of integration and testing, confirming that all components from the previous steps form the essential baseline for the newly integrated feature.

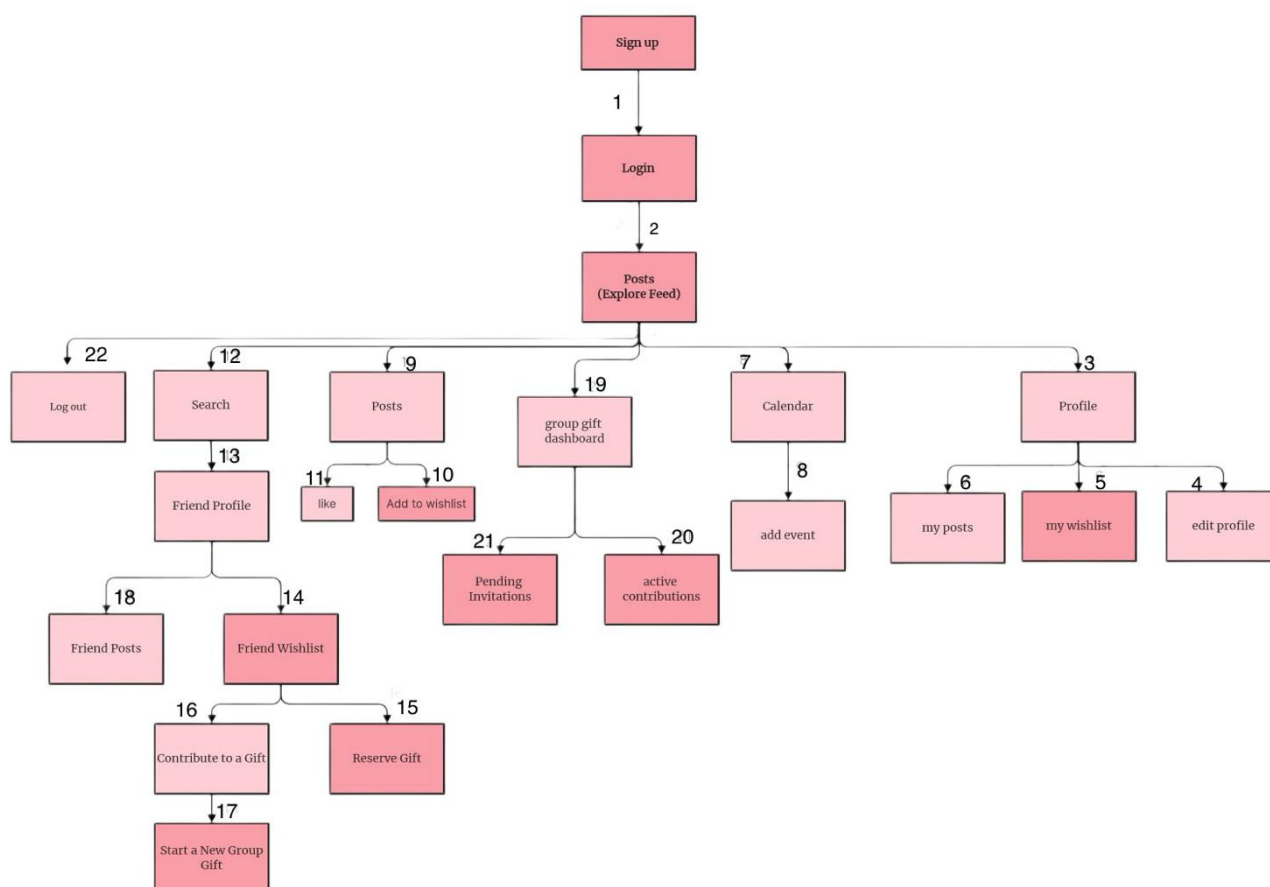


Figure10: Integration Hierarchy

5.2.2 integration testing plan

In this sub-section, we present the integration testing plan table, which aims to ensure that system components interact correctly with newly added components. The table includes relevant test cases along with an indication of whether each component passed the corresponding tests

System components	New component	Test cases	Pass?	Comments
- Register to the System	Log in to the System	1- A user registered a new account successfully. 2- The user then logged in to the system using the same credentials.	Yes	

- Register to the System - Log in to the System	Posts (Explore Feed)	1- A user registered a new account successfully. 2- The user then logged in to the system using the same credentials. 3. The Posts (Explore Feed) loads instantly, displaying a stream of gift ideas. 4. User is able to browse content freely and view posts.	Yes	
-Register to the System ‘ Log in to the System - Posts (Explore Feed)	Profile (My Wishlist, My Posts, Edit Profile)	1- A user registered a new account successfully. 2- The user then logged in to the system using the same credentials. 3. The Posts (Explore Feed) loads instantly, displaying a stream of gift ideas. 4. User is able to browse content freely and view posts. 5. Navigate from Posts (Explore Feed) to Profile. 6. Verify Profile loads successfully. 7. Verify My Wishlist and My Posts displays load correctly (Read function confirmed). 8. Use Edit Profile to update bio/info and save (Update function confirmed). 9. System confirms save, and updated info appears on Profile view.	Yes	

-Register to the System ‘ Log in to the System -Posts (Explore Feed) -Profile (My Wishlist, My Posts, Edit Profile)	Calendar, Add Event	1- A user registered a new account successfully. 2- The user then logged in to the system using the same credentials. 3. The Posts (Explore Feed) loads instantly, displaying a stream of gift ideas. 4. User is able to browse content freely and view posts. 5. Navigate from Posts (Explore Feed) to Profile. 6. Verify Profile loads successfully. 7. Verify My Wishlist and My Posts displays load correctly (Read function confirmed). 8. Use Edit Profile to update bio/info and save (Update function confirmed). 9. System confirms save, and updated info appears on Profile view. 10. User navigates from Profile to Calendar. 11. User successfully uses Add Event to create a new occasion (e.g., "Birthday"). 12. The new event is displayed correctly on the Calendar view.	Yes	
-Register to the System ‘ Log in to the System ‘	Like Post, Add to Wishlist	1- A user registered a new account successfully. 2- The user then logged in to the system using	Yes	

-Posts (Explore Feed) -Profile (My Wishlist, My Posts, Edit Profile) -Calendar, Add Event		the same credentials. 3. The Posts (Explore Feed) loads instantly, displaying a stream of gift ideas. 4. User is able to browse content freely and view posts. 5. Navigate from Posts (Explore Feed) to Profile. 6. Verify Profile loads successfully. 7. Verify My Wishlist and My Posts displays load correctly (Read function confirmed). 8. Use Edit Profile to update bio/info and save (Update function confirmed). 9. System confirms save, and updated info appears on Profile view. 10. User navigates from Profile to Calendar. 11. User successfully uses Add Event to create a new occasion (e.g., "Birthday"). 12. The new event is displayed correctly on the Calendar view. 13. User navigates back to Posts (Explore Feed). 14. User clicks Like on a post. The system verifies the 'Like' count updates correctly. 15. User clicks "Add to Wishlist". 16. User verifies the new item appears in My		
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		Wishlist (via Profile).		
-Register to the System ‘ Log in to the System - Posts (Explore Feed) -Profile (My Wishlist, My Posts, Edit Profile) -Calendar, Add Event -Like Post, Add to Wishlist	Search for Friends, View Friend Profile, View Friend Posts, View Friend Wishlist, Contribute to a Gift, Start a New Group Gift, Reserve Gift	1- A user registered a new account successfully. 2- The user then logged in to the system using the same credentials. 3. The Posts (Explore Feed) loads instantly, displaying a stream of gift ideas. 4. User is able to browse content freely and view posts. 5. Navigate from Posts (Explore Feed) to Profile. 6. Verify Profile loads successfully. 7. Verify My Wishlist and My Posts displays load correctly (Read function confirmed). 8. Use Edit Profile to update bio/info and save (Update function confirmed). 9. System confirms save, and updated info appears on Profile view. 10. User navigates from Profile to Calendar. 11. User successfully uses Add Event to create a new occasion (e.g., "Birthday"). 12. The new event is displayed correctly on the Calendar view. 13. User navigates back to Posts (Explore Feed). 14. User clicks Like on	Yes	

		a post. The system verifies the 'Like' count updates correctly. 15. User clicks "Add to Wishlist". 16. User verifies the new item appears in My Wishlist (via Profile). 17. User accesses the Search feature. 18. User successfully finds and loads a Friend Profile. 19. User verifies Friend Posts loads correctly. 20. User accesses Friend Wishlist. 21. User clicks Contribute to a Gift / Start a New Group Gift and successfully initiates the group process. 22. User clicks Reserve Gift on a different item, and the item status successfully updates to 'Reserved' (Visible to all viewers)		
-Register to the System ‘ Log in to the System - Posts (Explore Feed) -Profile (My Wishlist, My Posts, Edit Profile) -Calendar, Add Event -Like Post, Add to Wishlist	Group Gift Dashboard (Pending Invitations, Active Contributions)	1- A user registered a new account successfully. 2- The user then logged in to the system using the same credentials. 3. The Posts (Explore Feed) loads instantly, displaying a stream of gift ideas. 4. User is able to browse content freely and view posts. 5. Navigate from Posts (Explore Feed) to Profile. 6. Verify Profile loads successfully. 7.	Yes	

-Search for Friends, View Friend Profile, View Friend Posts, View Friend Wishlist, Reserve Gift		Verify My Wishlist and My Posts displays load correctly (Read function confirmed). 8. Use Edit Profile to update bio/info and save (Update function confirmed). 9. System confirms save, and updated info appears on Profile view. 10. User navigates from Profile to Calendar. 11. User successfully uses Add Event to create a new occasion (e.g., "Birthday"). 12. The new event is displayed correctly on the Calendar view. 13. User navigates back to Posts (Explore Feed). 14. User clicks Like on a post. The system verifies the 'Like' count updates correctly. 15. User clicks "Add to Wishlist". 16. User verifies the new item appears in My Wishlist (via Profile). 17. User accesses the Search feature. 18. User successfully finds and loads a Friend Profile. 19. User verifies Friend Posts loads correctly. 20. User accesses Friend Wishlist. 21. User clicks Contribute to a Gift / Start a New Group Gift and successfully initiates the group		
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		process. 22. User clicks Reserve Gift on a different item, and the item status successfully updates to 'Reserved' (Visible to all viewers 23. User navigates to the Group Gift Dashboard (which branches from Posts). 24. The Active Contributions section displays the status of the previously initiated Group Gift correctly. 25. The Pending Invitations section loads correctly, showing any invitations sent or received.		
-Register to the System ‘ Log in to the System - Posts (Explore Feed) -Profile (My Wishlist, My Posts, Edit Profile) -Calendar, Add Event -Like Post, Add to Wishlist -Search for Friends, View Friend Profile, View Friend Posts, View Friend Wishlist, Reserve Gift	Log Out	1- A user registered a new account successfully. 2- The user then logged in to the system using the same credentials. 3. The Posts (Explore Feed) loads instantly, displaying a stream of gift ideas. 4. User is able to browse content freely and view posts. 5. Navigate from Posts (Explore Feed) to Profile. 6. Verify Profile loads successfully. 7. Verify My Wishlist and My Posts displays load correctly (Read function confirmed). 8. Use Edit Profile to update bio/info and save (Update	Yes	

-Group Gift Dashboard (Pending Invitations, Active Contributions)		function confirmed). 9. System confirms save, and updated info appears on Profile view. 10. User navigates from Profile to Calendar. 11. User successfully uses Add Event to create a new occasion (e.g., "Birthday"). 12. The new event is displayed correctly on the Calendar view. 13. User navigates back to Posts (Explore Feed). 14. User clicks Like on a post. The system verifies the 'Like' count updates correctly. 15. User clicks "Add to Wishlist". 16. User verifies the new item appears in My Wishlist (via Profile). 17. User accesses the Search feature. 18. User successfully finds and loads a Friend Profile. 19. User verifies Friend Posts loads correctly. 20. User accesses Friend Wishlist. 21. User clicks Contribute to a Gift / Start a New Group Gift and successfully initiates the group process. 22. User clicks Reserve Gift on a different item, and the item status successfully updates to 'Reserved' (Visible to all viewers		
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		23. User navigates to the Group Gift Dashboard (which branches from Posts). 24. The Active Contributions section displays the status of the previously initiated Group Gift correctly. 25. The Pending Invitations section loads correctly, showing any invitations sent or received. 26. User performs actions across all integrated modules (from Login to viewing the Dashboard).27. User clicks Log Out from the navigation menu. 28. System terminates the session and redirects the user to the Welcome/Login page.		
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Table 9: Integration Testing Plan

5.3 User Acceptance Testing

User Acceptance Testing (UAT) is the final stage of evaluation, where actual users from the target audience interact with the system in real-life scenarios. This type of testing helps verify that the system functions as intended, meets user needs, and is ready for deployment. To conduct this test, we recruited 10 participants who matched the characteristics of our target audience, as defined in Phase 2. The participants' ages ranged from 18 to 35 years.

At the beginning of each session, participants were given a consent form to provide their personal information and confirm their agreement to participate (See Appendix D for the Participant Consent Form).

We then provided them with a scenario that covered the core functionalities of the WOD system, including signing up, creating a post, managing a wishlist, reserving a gift, contributing to a group gift, and adding an occasion to the calendar. Before starting the timed tasks, participants were given a few minutes to explore the system freely. Afterwards, they followed the provided scenario to complete the specific tasks. Their feedback was gathered through a post-test questionnaire consisting of 10 questions (See Appendix E for the User Acceptance Testing Questionnaire), where they

were asked to evaluate their experience, report any issues, and suggest improvements. The responses were later analyzed to identify usability problems and guide future enhancements.

5.3.1 Demographics of Participants

In this sub-section, we present the participants' demographic information, which includes data regarding their age, gender, technical experience, and whether they have used similar gifting products before. The demographic results are summarized in the following pie chart:

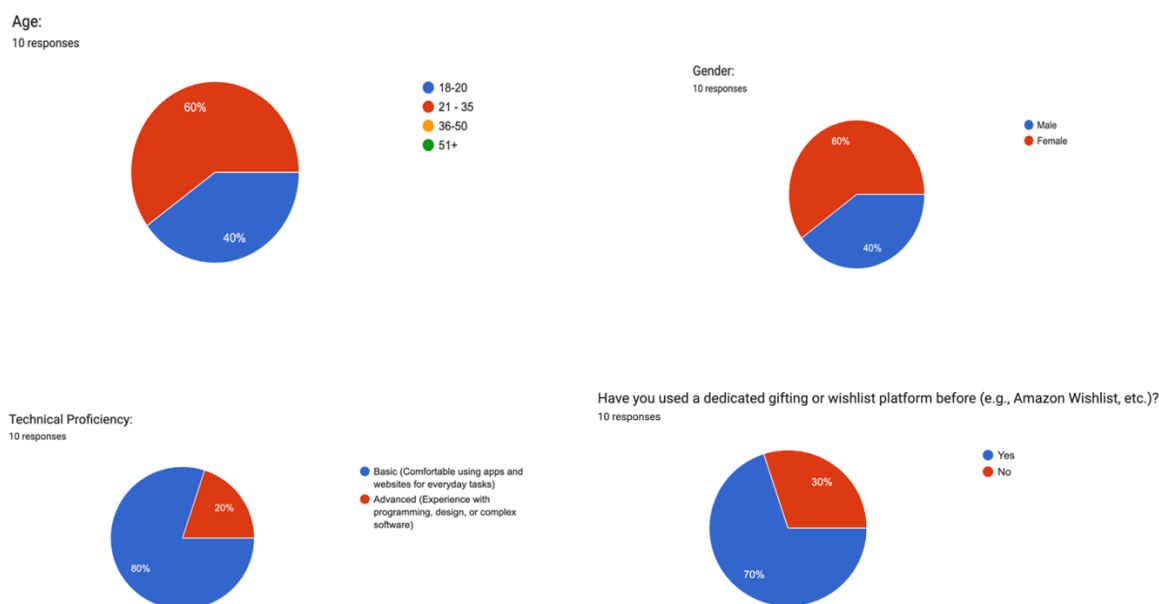


Figure 10: Demographics of Acceptance Testing Participants

Age Groups: All participants were young adults, with 60% falling in the 21-35 age bracket and 40% being between 18 and 20 years old. This aligns well with our target demographic of younger users who are typically more engaged with social and gifting technologies.

Gender Distribution: The participant group was composed of 60% female and 40% male users. This balanced gender distribution allows for diverse perspectives in the user feedback.

Technical Experience: A significant majority of participants (80%) reported having a basic level of technical proficiency, being comfortable with everyday apps and websites. The remaining 20% identified as having advanced technical experience. This distribution is valuable for assessing the platform's intuitiveness and learning curve for non-technical users.

Previous Experience with Similar Products: Unlike our initial expectations, most participants (70%) indicated that they had prior experience with a dedicated gifting or wishlist platform. The remaining 30% had no such experience. This provides an excellent opportunity to gather feedback from users who can compare our platform's unique features, like group contribution and gift reservation, against existing solutions.

5.3.2 Questionnaire Result

In this sub-section, a questionnaire (see Appendix E) was distributed to the 10 users to evaluate the website's usability, performance, and overall user satisfaction. The survey collected detailed feedback on specific tasks and perceptions. The subsequent analysis summarizes and interprets the quantitative results and qualitative feedback obtained from the participants.

Task Efficiency and Core Feature Usability: The data indicates a high level of user proficiency with the platform's primary functions. The account registration and login process was found to be straightforward, with all participants (100%) agreeing or strongly agreeing. Furthermore, every participant (100%) strongly agreed that the process of creating a post and adding it to a wishlist was simple and understandable, confirming the intuitiveness of this fundamental feature. Navigation between different website sections was also highly rated, with 100% of users providing positive feedback (70% strongly agreeing, 30% agreeing).

Validation of Key Innovations: The unique features designed to address core user pain points were overwhelmingly validated. The gift reservation feature was unanimously perceived as clear and effective, with 100% of participants strongly agreeing it would prevent duplicate gifts. Similarly, the group gifting feature was received very positively, with all participants (100%) agreeing it would simplify coordination, 80% of them strongly so.

User Experience and Perception: Users reported a strong sense of clarity and control while using the platform. A majority (90%) found the website's interface clear and easy to understand, with 80% strongly agreeing. This sentiment extended to users' confidence, as all participants (100%) felt confident and in control of the gift planning process, with 80% strongly agreeing. When searching for specific information, such as a friend's profile or a gift item, all participants (100%) reported success, with 80% strongly agreeing on the ease of finding information.

Overall Satisfaction and Performance: The overall response to the platform was highly positive. When asked about their overall satisfaction and likelihood to recommend the WOD platform, 90% of users responded positively (80% strongly agreeing, 10% agreeing), while 10% were neutral. In terms of technical performance, the website's response speed was rated positively by 90% of users, with 70% rating it as excellent (5) and 20% as good (4). One user (10%) experienced the speed as average (3), indicating a generally strong performance with minor room for optimization.

Areas for Improvement: While the quantitative feedback was overwhelmingly positive, qualitative suggestions from participants provided valuable direction for future iterations. The most common suggestions included:

- Implementing push notification reminders for upcoming occasions.
 - Developing a more advanced search and filter system for the Explore feed.
 - Introducing the ability to create private wishlists for specific events (e.g., a wedding registry).
- These insights highlight concrete opportunities to enhance usability and increase user engagement in subsequent development cycles.

5.4 Discussion

This section provides an interpretation of the results obtained during the testing phases and summarizes reflections on the development process, system design, and evaluation outcomes.

Process Evaluation: The development process went through structured phases, including interface design, database setup, feature implementation, and continuous testing.

User Story Acceptance Testing ensured each function (signup, login, wishlist visibility, posting, following, searching, calendar use) met its intended requirement.

Integration Testing confirmed that the features worked correctly together without conflicts.

Overall, the testing methods were effective, and most functionalities behaved as expected. Minor issues—mainly layout inconsistencies—were identified and resolved throughout the process.

Design and Development Reflections: The system’s design evolved significantly during development. The platform became more consistent, visually appealing, and easier to navigate, especially after unifying styles across pages, fixing alignment issues, and improving button and icon placement. Some challenges encountered included:

- Ensuring consistent styling across different pages
- Fixing UI bugs caused by CSS conflicts
- Adjusting visual elements to work on all screen sizes
- Ensuring smooth interaction with the database for features like wishlist visibility and follow/unfollow

Despite these issues, the final interface is clean, intuitive, and user-friendly.

Results Interpretation: User Acceptance Testing produced mostly positive feedback.

Users were able to:

- Create posts easily
- Add items to their wishlist
- Follow and unfollow users
- Search for friends
- Use the calendar to view gift dates and events

Participants found the system simple and pleasant to use.

Some visual suggestions were made—mainly related to spacing, alignment, and color contrast—but no major usability problems were reported.

Areas for Improvement:

Based on the results and feedback, the following improvements can enhance the system:

- **UI Consistency:** Some pages still require better color and typography consistency.
- **Performance:** Image-heavy pages could be optimized to load faster.
- **More Features:** Filters, sorting options, or personalized feeds could enrich the experience.
- **User Guidance:** Certain features, such as wishlist visibility, could include clearer instructions or tooltips.

6 Chapter 6: Conclusion and Future Work

6.1 Conclusion

In conclusion, this project successfully delivers a social platform that allows users to share wishlist items, create posts, follow friends, and organize gift-related events through a personalized calendar. It solves the common problem of keeping track of personal preferences and gift ideas in a simple and structured way.

This report covered the full development lifecycle, including:

- Problem definition and proposed solution
- Requirements analysis
- System design and architecture
- Database structure and feature implementation
- User interface development
- System testing (UAT, integration, and non-functional testing)

The results demonstrated that the platform is functional, intuitive, and aligned with user needs. The system meets its main objective by providing a fun, organized, and visually appealing space for sharing gift ideas and personal preferences.

6.2 Global and Local Impact

- **Local Impact:** Helps families, friends, and students stay organized with events and gift preferences.
- **Global Impact:** Has the potential to grow into a broader social platform focused on gifting, recommendations, and wishlists.

6.3 Problems and Challenges

Throughout development, several challenges were encountered:

- CSS conflicts causing layout misalignment
- Maintaining a consistent design across all pages
- Ensuring buttons and icons behaved correctly in all cases
- Handling database updates for dynamic functions like visibility toggling
- Differences in appearance between desktop and mobile views

These challenges shaped the refinement process and strengthened the final design.

6.4 Limitations of the System

Current limitations include:

- No mobile application version
- No notification system
- No chat or messaging features
- Some pages rely on large images, which may slow loading
- No AI recommendations or advanced sorting features yet

6.5 Main Contribution

The project's key contributions include:

- A functional and visually cohesive social gifting platform
- A complete wishlist system with privacy controls
- A calendar for tracking events and gift ideas
- A social element enabling users to follow each other
- A solid foundation for future enhancements

6.6 Future Work

Future improvements can significantly extend system capabilities:

1. **Mobile App Development** (iOS + Android)
2. **Notification System** (wishlist updates, new followers, upcoming events)
3. **Filtering and Sorting Tools** for posts and wishlists
4. **Built-in Chat System** to enhance social interaction
5. **Calendar Enhancements** with reminders and auto-notifications
6. **AI-powered Gift Recommendations**
7. **Video support** for posts
8. **Improved Security Features**

7 References

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8 Appendix

8.1 Appendix_A: Interview

8.1.1 Interview questions:

1. Tell me about the last time you bought a gift for a friend or relative. What was that process like?
2. Have you ever purchased a gift, only to discover the recipient already received the exact same item from someone else? If so, could you describe what happened, how that made you feel, and whether you wished there was a simple way to prevent that duplication?
3. How do you currently ensure you remember all the important birthdays and occasions for your friends and family?"
4. When you coordinate with a group to collect money for a gift, what is the most annoying part of the coordination process (collecting money, making the decision, etc.)?
5. When buying a gift, would you prefer that the recipient have a clear wishlist available, or do you prefer to guess and choose a surprise? Could you explain why?
6. When you need to buy a specific local gift (like a custom cake or flowers), what are the three most time-consuming steps you go through right now? Would seeing local vendor catalogs directly within an app eliminate those steps for you?
7. Is there anything else you would like to add about the gift-giving experience that you struggle with ?

8.1.2 Interview transcriptions

The following are the transcripts that represent the full five interviews we conducted during the requirements elicitation process, including interviewee's backgrounds, interview objectives and our observations.

Interview#1	
Interviewee: Hadeel ibrahim	Interviewer: Reema Alraqibah
Location/ Medium: Face to face in Hadeel's house, Riyadh, Saudi Arabia	Appointment Date: 3 october 2025 Start Time: 16:15 End Time: 16:55
Objectives • Validate Duplication & Financial Stress: To assess the direct impact of gift duplication and identify the highest financial friction point in group coordination. • Determine Utility vs. Habit: To evaluate user reliance on automated reminders and gauge the perceived value of the Wishlist feature as a tool for efficiency and guaranteed satisfaction. • Identify Inspiration & Logistical Needs: To pinpoint key struggles related to finding unique gift ideas and securing simple, timely delivery logistics.	Reminders: Was blocked from sending gifts outside her city (The Chefz issue). Export to Sheets
Agenda: Introduction Background in project Overview of interview Topics to be covered Question 1 Question 2 Question 3 Question 4 Question 5 Question 6 Question 7 Summary of major points Questions from interviewee Closing	Approximate Time: 3 minutes 3 minutes 2 minutes 2 minutes 4 minutes 2 minutes 2 minutes 5 minutes 4 minutes 6 minutes 3 minutes 2 minutes 1 minutes 2minutes

Table 8: Interview's Transcription for first Interviewee

Interview#1
General Observations: The interviewee shows high sensitivity to logistical roadblocks (e.g., city restrictions in apps) and values presentation quality (professional wrapping). She views price comparison and finding external shops as major time-consuming hurdles, despite her personal preference against a general wishlist.

Topic not covered: The technical requirements for implementing a custom link/post interface were not discussed in detail, as the focus remained on the user's current frustrations with external apps. Interviewee: Hadeel Ibrahim Date: 3 October 2025	
Q1. Tell me about the last time you bought a gift for a friend or relative. What was that process like?	Answer: On my friend's birthday, I was outside the city where she lives. The app "the Shifz" wouldn't let me send a gift since I wasn't in the same city, so I had to use another app with higher prices just to make sure the gift reached her. Observation: The user faces critical logistical and geographic constraints with current gifting apps, often leading to increased cost and effort. This validates the need for a flexible, referral-based system.
Q2. Have you ever purchased a gift, only to discover the recipient already received the exact same item from someone else? If so, could you describe what happened, how that made you feel, and whether you wished there was a simple way to prevent that duplication?	Answer: No, that never happened to me. Observation: She has not personally experienced duplication.
Q3. How do you currently ensure you remember all the important birthdays and occasions for your friends and family?	Answer: I usually depend on my memory, but sometimes I forget. Occasionally, Snapchat reminds me of birthdays. Observation: Her primary reminder mechanism (personal memory + unreliable social media) is prone to failure, supporting the demand for a reliable, dedicated calendar/reminder system.
Q4. When you coordinate with a group to collect money for a gift, what is the most annoying part of the coordination process (collecting money, making the decision, etc.)?	Answer: The main challenge is choosing the right gift, since not everyone in the group agrees because of different budgets. Observation: Her main pain point in coordination is budget disparity leading to decision paralysis, rather than just collecting money. This emphasizes the need for a feature that facilitates budget agreement.
Q5. When buying a gift, would you prefer that the recipient have a clear Wishlist available, or do you prefer to guess and choose a surprise? Could you explain why?	Answer: No, because the gift has to carry my personal touch so that they remember me through it. It's not just about what they want or need. Observation: Clear preference for the social/emotional value (surprise, personal stamp) over sheer utility (guaranteed satisfaction), highlighting that the Wishlist must remain optional and flexible.
Q6. When you need to buy a specific local gift (like a custom cake or flowers), what are the three most	Answer:

time-consuming steps you go through right now? Would seeing local vendor catalogs directly within an app eliminate those steps for you?	The most time-consuming steps are: getting the flowers wrapped professionally and attractively, choosing the cake with the right flavors, and comparing prices. Observation: User prioritizes Presentation Quality (wrapping) and accurate information (flavors, comparison), suggesting the visual inspiration feed must prioritize high-quality images and clear descriptions.
Q7. Is there anything else you would like to add about the gift-giving experience that you struggle with?	Answer: Sometimes I know exactly what gift I want to buy, but I don't know the right place to get it from Observation: This directly validates the core value of the new Explore/Post feature —to connect a specific, desired idea (the inspiration) with a link to its source (the solution).

Table 9: Another Interview's Transcription for first Interviewee.

Interview#2	
Interviewee: Joud Abdullah	Interviewer: shoog Alawdah
Location/ Medium: phone call	Appointment Date: 4 October 2025 Start Time: 18:00 End Time: 18:40
Objectives: Validate Duplication & Financial Stress: To assess the direct impact of gift duplication and identify the highest financial friction point in group coordination. Determine Utility vs. Habit: To evaluate user reliance on automated reminders and gauge the perceived value of the Wishlist feature as a tool for efficiency and guaranteed satisfaction. Identify Inspiration & Logistical Needs: To pinpoint key struggles related to finding unique gift ideas and securing simple, timely delivery logistics.	Reminders: Critical Pain Point: Experienced costly gift duplication (baby stroller) and emphasizes the need for coordination. Coordination Friction: Money collection/transfer is highly inconvenient and suffers from delayed payments. Utility Focus: Strongly prefers the Wishlist as it saves time and guarantees the recipient's satisfaction.
Agenda: Introduction Background in project Overview of interview Topics to be covered Question 1 Question 2	Approximate Time: 3 minutes 3 minutes 2 minutes 2 minutes 4 minutes 2 minutes 2 minutes

Question 3	5 minutes
Question 4	4 minutes
Question 5	6 minutes
Question 6	3 minutes
Question 7	2 minutes
Summary of major points	1 minutes
Questions from interviewee	2minutes
Closing	

Table 9: Interview's Transcription for second Interviewee.

Interview#2	
General Observations: The interviewee strongly validated the critical pain points of duplication, describing the experience as emotionally and financially upsetting. She exhibits a clear preference for utility and efficiency (Wishlist) over the element of surprise, and pinpoints delayed payment and coordination confusion as the major friction points in group gifting Topic not covered: The specific technical details or methods for securely transferring/collecting funds within the group contribution feature were not explored in depth. Interviewee: Joud Abdullah Date: 4 October 2025	
Q1. Tell me about the last time you bought a gift for a friend or relative. What was that process like?	Answer: On a friend's birthday, I bought her a gift from Sephora, but I was worried she might not like it or that she already had the same thing. Observation: The participant suffers from anxiety regarding duplication and recipient satisfaction, validating the need for a mechanism (like a Wishlist or reservation) to mitigate this risk.
Q2. Have you ever purchased a gift, only to discover the recipient already received the exact same item from someone else? If so, could you describe what happened, how that made you feel, and whether you wished there was a simple way to prevent that duplication?	Answer: Yes, it happened at a baby shower for a relative. I bought an expensive baby stroller, and when I got there, I found out someone else had brought the exact same stroller. I was upset because it cost me a lot, and there was nothing I could do. I really wished there had been some kind of coordination among the guests to avoid duplicate gifts. Observation: Confirms the significant financial and emotional impact of duplication. This is crucial evidence supporting the Gift Reservation feature's high priority.
Q3. How do you currently ensure you remember all the important birthdays and occasions for your friends and family?"	Answer: I only remember a few occasions for the people closest to me, but I often forget the rest. This really bothers me because it makes me feel embarrassed and unprepared when it comes to choosing the right gift. Observation:

	Reliance on memory leads to social failure and emotional distress . This strongly validates the necessity for the Calendar and Reminder System .
Q4. When you coordinate with a group to collect money for a gift, what is the most annoying part of the coordination process (collecting money, making the decision, etc.)?	Answer: Sometimes not all group members are available during the discussion... Also, transferring money between us can be inconvenient, and sometimes people delay payments. Observation: The primary obstacle is financial/logistical friction (delayed payments and inconvenient transfers), alongside the communication hurdle of incomplete attendance.
Q5. When buying a gift, would you prefer that the recipient have a clear wishlist available, or do you prefer to guess and choose a surprise? Could you explain why?	Answer: I prefer that the recipient has a ready wishlist, so I can choose something directly from it. This saves me time, especially if it's someone I don't know very well. Observation: Strong preference for efficiency and utility. She views the Wishlist as a time-saving, guaranteed satisfaction tool, validating its role as a core feature.
Q6. When you need to buy a specific local gift (like a custom cake or flowers), what are the three most time-consuming steps you go through right now? Would seeing local vendor catalogs directly within an app eliminate those steps for you?	Answer: The time-consuming parts are: choosing the right cake flavor, selecting the design, and dealing with possible allergies... Also, I can't always find shops that allow me to customize the design of chocolates. Observation: Reveals a demand for both detailed product information (allergies, flavors) and the ability to find customized or unique items (designing own chocolates) via the Explore Feed.
Q7. Is there anything else you would like to add about the gift-giving experience that you struggle with ?	Answer: Coordinating gifts among friends and agreeing on a specific budget would be much better. It avoids awkward situations when one person buys a gift worth 1000 riyals while another just brings chocolates worth 50 riyals. Observation: Highlights a need for a budget negotiation mechanism or social tool to mitigate the social awkwardness arising from widely varying contribution values.

Table 10: Another Interview's Transcription for second Interviewee

Interview#3	
Interviewee: Reema Adel	Interviewer: Noura Alamro
Location/ Medium: phone call	Appointment Date: 7 October 2025 Start Time: 22:00 End Time: 22:35
Objectives:	Reminders:

1. Validate Duplication & Financial Stress: To assess the direct impact of gift duplication and identify the highest financial friction point in group coordination. 2. Determine Utility vs. Habit: To evaluate user reliance on automated reminders and gauge the perceived value of the Wishlist feature as a tool for efficiency and guaranteed satisfaction. 3. Identify Inspiration & Logistical Needs: To pinpoint key struggles related to finding unique gift ideas and securing simple, timely delivery logistics.	Interviewee enjoys thoughtful and creative gifting experiences. She often takes part in social events and celebrations, which makes her a suitable candidate to discuss gift-giving habits and challenges.
Agenda: Introduction Background in project Overview of interview Topics to be covered Question 1 Question 2 Question 3 Question 4 Question 5 Question 6 Question 7 Summary of major points Questions from interviewee Closing	Approximate Time: 3 minutes 2 minutes 2 minutes 1 minutes 2 minutes 3 minutes 4 minutes 2 minutes 3 minutes 3 minutes 5 minutes 2 minutes 1 minutes 2minutes

Table 11: Interview's Transcription for third Interviewee.

Interview#3	
General Observations: Reema values meaningful and personalized gifts that reflect thoughtfulness. She prefers giving surprises rather than relying on wishlists and tends to rely on others for reminders about special occasions. While she enjoys the gifting process, she struggles with time-consuming tasks like comparing prices and finding reliable stores for local gifts. Topic not covered: Due to time constraints, we couldn't further explore the interviewee's suggestions or ideas for improving the gift-giving experience. Interviewee: Reema Adel Date: 7 October 2025	
Q1. Tell me about the last time you bought a gift for a friend or relative. What was that process like?	Answer: I bought my cousin a pair of earrings for her graduation. She had been wanting them for a while since she already had the matching necklace, so I thought it would be nice to complete the set for her.

	Observation: The participant chooses gifts that carry sentimental value and reflect personal connection and attention to detail.
Q2. Have you ever purchased a gift, only to discover the recipient already received the exact same item from someone else? If so, could you describe what happened, how that made you feel, and whether you wished there was a simple way to prevent that duplication?	Answer: No, that never happened to me. Observation: The participant likely takes care to avoid duplication.
Q3. How do you currently ensure you remember all the important birthdays and occasions for your friends and family?"	Answer: Honestly, I just rely on people to remind me about important occasions. I don't really keep track myself. Observation: They don't use organized reminder tools, indicating potential value for automated reminder features.
Q4. When you coordinate with a group to collect money for a gift, what is the most annoying part of the coordination process (collecting money, making the decision, etc.)?	Answer: The hardest part is when we can't agree on what gift to get. Everyone has different opinions, and it can get a little frustrating. Observation: Decision-making within groups is stressful, suggesting that features simplifying group coordination could be valuable.
Q5. When buying a gift, would you prefer that the recipient have a clear wishlist available, or do you prefer to guess and choose a surprise? Could you explain why?	Answer: I prefer to guess and surprise the person. I think it's more fun that way, and it feels more personal than just picking something from a wishlist. Observation: They value emotional and personal aspects of gifting more than convenience, indicating personalization is important.
Q6. When you need to buy a specific local gift (like a custom cake or flowers), what are the three most time-consuming steps you go through right now? Would seeing local vendor catalogs directly within an app eliminate those steps for you?	Answer: The most time-consuming part for me is figuring out what type of gift would really suit the person. I often have to switch between multiple apps or websites to get ideas, which takes a lot of time. I also prefer to see real photos or examples of gifts rather than just reading plain text lists, because it helps me visualize the gift and know if it feels personal and thoughtful. Observation: Reema struggles with scattered sources and lack of visual inspiration, highlighting the need for a centralized, image-based platform that makes finding meaningful and personalized gift ideas faster and easier.
Q7. Is there anything else you would like to add about the gift-giving experience that you struggle with ?	Answer: I like to know what other people are bringing so the gifts don't end up being the same. It just makes things smoother. Observation:

	They prefer better coordination among gift-givers, reinforcing the idea of shared visibility or collaborative planning features.
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Table 12: Another Interview's Transcription for third Interviewee.

Interview#4	
Interviewee: Najoud AlQahtani	Interviewer: Ghada AlQahtani
Location/ Medium: phone call	Appointment Date: 5 October 2025 Start Time: 16:00 End Time:16:40
Objectives: Validate Duplication & Financial Stress: To assess the direct impact of gift duplication and identify the highest financial friction point in group coordination. Determine Utility vs. Habit: To evaluate user reliance on automated reminders and gauge the perceived value of the Wishlist feature as a tool for efficiency and guaranteed satisfaction. Identify Inspiration & Logistical Needs: To pinpoint key struggles related to finding unique gift ideas and securing simple, timely delivery logistics.	Reminders: The gift-buying process is often time-consuming, especially when comparing prices or visiting multiple stores. • Duplicate gifts can cause disappointment, showing a need for a system that prevents such situations. • Most users rely on phone calendars to remember occasions, but would prefer a smarter, automated reminder system. • Coordinating group gifts is stressful, mainly due to collecting money and differing opinions. • Users prefer having access to wishlists rather than guessing what to buy, though they still like to add a personal touch. • There's a clear interest in discovering unique and personalized gift ideas that match the recipient's preferences.
Agenda: Introduction Background in project Overview of interview Topics to be covered Question 1 Question 2 Question 3 Question 4 Question 5 Question 6 Question 7 Summary of major points Questions from interviewee Closing	Approximate Time: 3 minutes 3 minutes 2 minutes 2 minutes 4 minutes 2 minutes 2 minutes 5 minutes 4 minutes 6 minutes 3 minutes 2 minutes 1 minutes 2minutes

Table 13: Interview's Transcription for fourth Interviewee

Interview#4	
General Observations: The interview showed that the gift-giving process is often time-consuming and stressful, with a need for easier ways to organize occasions, avoid duplicates, and manage group gifts Topic not covered: Privacy concerns and sharing wishlists were not discussed Interviewee: Najoud AlQahtani Date: 5 October 2025	
Q1. Tell me about the last time you bought a gift for a friend or relative. What was that process like?	Answer: It was a graduation gift for my friend. I spent a long time searching online, comparing prices, and then went to buy it from a store. Honestly, it took quite a bit of time and effort. Observation: The process of buying gifts takes a lot of time and effort, showing a need for an easier and faster way to find suitable gifts
Q2. Have you ever purchased a gift, only to discover the recipient already received the exact same item from someone else? If so, could you describe what happened, how that made you feel, and whether you wished there was a simple way to prevent that duplication?	Answer: Yes, once I bought a perfume for a friend, and she had already received the same one. It was disappointing, and I wished there was a way to avoid that. Observation: Gift duplication causes disappointment, indicating the need for a system to prevent buying the same gifts
Q3. How do you currently ensure you remember all the important birthdays and occasions for your friends and family?"	Answer: I save the dates in my phone calendar and set reminders a few days in advance so I can prepare the gift on time. Observation: Using phone reminders helps but requires manual effort; an automatic reminder system would be more effective.
Q4. When you coordinate with a group to collect money for a gift, what is the most annoying part of the coordination process (collecting money, making the decision, etc.)?	Answer: Collecting the money from everyone is the hardest part. Some people delay or change their mind about the gift, which makes the process stressful. Observation: Group gift coordination is stressful, showing a need for a simpler and more organized way to manage contributions.
Q5. When buying a gift, would you prefer that the recipient have a clear wishlist available, or	Answer: I prefer when the recipient has a wishlist, it's easier and ensures they'll like the gift. But

do you prefer to guess and choose a surprise? Could you explain why?	sometimes, I still like to add a small surprise to make it more personal. Observation: Having a wishlist helps choose gifts that suit the recipient while still allowing a personal touch.
Q6. When you need to buy a specific local gift (like a custom cake or flowers), what are the three most time-consuming steps you go through right now? Would seeing local vendor catalogs directly within an app eliminate those steps for you?	Answer: Finding good stores, checking their options, and contacting them usually take the most time. Having all local vendors in one app would make it much faster and easier. Observation: Searching and takes time; combining vendors in one app would simplify the process.
Q7. Is there anything else you would like to add about the gift-giving experience that you struggle with ?	Answer: I wish there was a simple way to discover unique or personalized gift ideas that actually match the person's interests. Observation: There is a clear interest in finding personalized and unique gift suggestions that match each person's taste.

Table 14: Another Interview's Transcription for fourth Interviewee.

Interview#5	
Interviewee: Yazed Abdullaziz	Interviewer: Renad Essa Alowais
Location/ Medium: phone call	Appointment Date: 4 October 2025 Start Time: 12:00 End Time:12:40
Objectives: 4. Validate Duplication & Financial Stress: To assess the direct impact of gift duplication and identify the highest financial friction point in group coordination. 5. Determine Utility vs. Habit: To evaluate user reliance on automated reminders and gauge the perceived value of the Wishlist feature as a tool for efficiency and guaranteed satisfaction. 6. Identify Inspiration & Logistical Needs: To pinpoint key struggles related to finding unique gift ideas and securing simple, timely delivery logistics.	Reminders: Planning a thoughtful and personalized gift makes the process smoother and more meaningful. • The participant has not experienced duplicate gifts but avoids that by planning carefully. • Dates for occasions are usually saved manually using calendars or notes. • Collecting money and agreeing on a group gift are the hardest parts of coordination. • Prefers to buy gifts based on what they already know the person wants instead of guessing. • Finding suitable stores, comparing prices, and arranging delivery are the most time-consuming steps when buying local gifts.

Agenda:	Approximate Time:
Introduction	3 minutes
Background in project	3 minutes
Overview of interview	2 minutes
Topics to be covered	2 minutes
Question 1	4 minutes
Question 2	2 minutes
Question 3	2 minutes
Question 4	5 minutes
Question 5	4 minutes
Question 6	6 minutes
Question 7	3 minutes
Summary of major points	2 minutes
Questions from interviewee	1 minutes
Closing	2minutes

Table 15: Interview's Transcription for fifth Interviewee.

Interview#5	
General Observations: The interview highlighted that the gift-giving process can be complicated due to time management, group coordination, and delivery arrangements. There's also a preference for comfort and organization rather than spontaneous or uncertain choices. Topic not covered: The interview did not explore how technology or mobile applications could specifically improve the overall gifting experience. Interviewee: Yazed Abdullaziz Date: 4 October 2025	
Q1. Tell me about the last time you bought a gift for a friend or relative. What was that process like?	Answer: The last time I bought a gift was for my friend who had just finished military training. I already knew what he wanted—a gaming PC—so my friend and I planned to surprise him with it. Observation: Shows a preference for thoughtful and planned gifts that match the recipient's known interests.
Q2. Have you ever purchased a gift, only to discover the recipient already received the exact same item from someone else? If so, could you describe what happened, how that made you feel, and whether you wished there was a simple way to prevent that duplication?	Answer: No, I usually choose my gifts carefully and make sure no one else has bought the same thing. Observation: Careful planning prevents duplication, showing awareness and attention to detail.
Q3. How do you currently ensure you remember all the important birthdays and occasions for your friends and family?"	Answer: I usually add the dates to reminders on the calendar or write them down in my notes.

	Observation: Relies on manual tracking methods, which could be improved with automated reminders.
Q4. When you coordinate with a group to collect money for a gift, what is the most annoying part of the coordination process (collecting money, making the decision, etc.)?	Answer: Collecting money from everyone is the hardest part. Some people delay or forget, and sometimes we disagree on the gift because everyone has different tastes. Observation: Group coordination remains a major difficulty due to payment delays and differing opinions.
Q5. When buying a gift, would you prefer that the recipient have a clear wishlist available, or do you prefer to guess and choose a surprise? Could you explain why?	Answer: I usually already know what the person wants, so I buy it directly. I don't like guessing because they might not like the gift. Observation: Prefers certainty and practicality over surprises, valuing confidence in the gift choice.
Q6. When you need to buy a specific local gift (like a custom cake or flowers), what are the three most time-consuming steps you go through right now? Would seeing local vendor catalogs directly within an app eliminate those steps for you?	Answer: Finding suitable stores, checking prices, and arranging for delivery take the most time. Having a store that manages all that would make it much easier. Observation: Time management is a key issue; integrated store services could simplify the process.
Q7. Is there anything else you would like to add about the gift-giving experience that you struggle with ?	Answer: I prefer having gifts delivered by a courier rather than giving them in person because I feel awkward about people's reactions. Observation: Prefers convenience and minimal social interaction in the gifting process, emphasizing comfort and practicality.

Table 16: Another Interview's Transcription for fifth Interviewee

8.2 Appendix_B: Questionnaire

Following are the full questions and responses collected through the online questionnaire conducted for the **WOD** project. The questionnaire was distributed to individuals within the Saudi community who actively participate in social and family gift-giving. It includes the complete set of quantitative and closed-ended answers that illustrate user behaviors, challenges, and preferences related to the gifting process. These responses were analyzed to support the findings discussed in the **Questionnaire section** and to validate the core requirements of the WOD platform.

Q1: I often find it difficult to come up with a meaningful gift idea for someone.

- 1 (Strongly Disagree)
- 2 (Disagree)
- 3 (Neutral)
- 4 (Agree)
- 5 (Strongly Agree)

Q2: Coordinating contributions and payments for a group gift is usually stressful or confusing.

- 1 (Strongly Disagree)
- 2 (Disagree)
- 3 (Neutral)
- 4 (Agree)
- 5 (Strongly Agree)

Q3: Sometimes I forget or almost forget an important occasion (like a friend's birthday or an anniversary).

- 1 (Strongly Disagree)
- 2 (Disagree)
- 3 (Neutral)
- 4 (Agree)
- 5 (Strongly Agree)

Q4: I wish there was an easier way to know what my friends actually want as a gift.

- 1 (Strongly Disagree)
- 2 (Disagree)
- 3 (Neutral)
- 4 (Agree)
- 5 (Strongly Agree)

Q5: What is the single biggest frustration you face when trying to coordinate a gift with a group of friends?

- Difficulty collecting financial contributions from everyone
- Unclear on who has paid or not paid
- Some friends delay or refuse to participate
- Disagreements about what gift to buy
- Others (please specify)

Q6: If you could add one more feature to a gifting platform to make your life easier, what would it be?

- Personalized wishlist for each user
- Integration with local stores for easier purchases
- Group contribution and payment tracking system
- Reminder notifications for important occasions
- Other (please specify)

8.2.1 Questionnaire Results:

Q1:

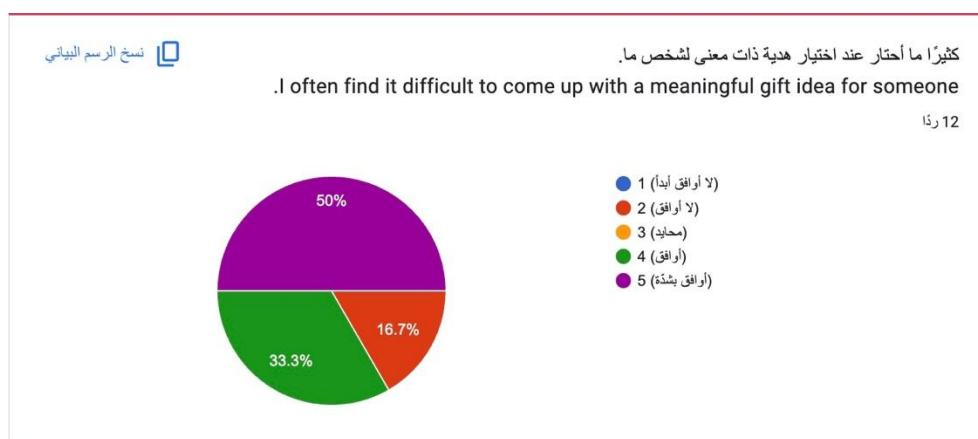


Figure 11: First question in questionnaire.

Q2:

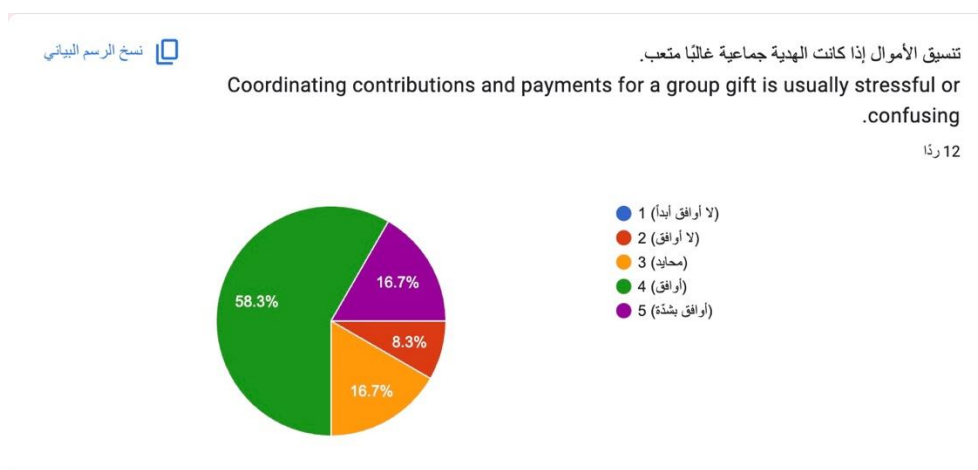


Figure 12: Second question in questionnaire.

Q3:



Figure 13: Third question in questionnaire

Q4:

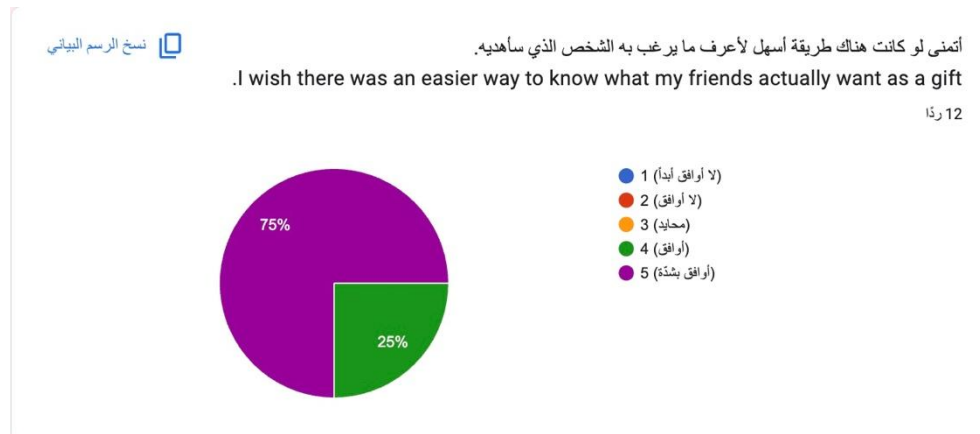


Figure 14: Fourth question in questionnaire

Q5:

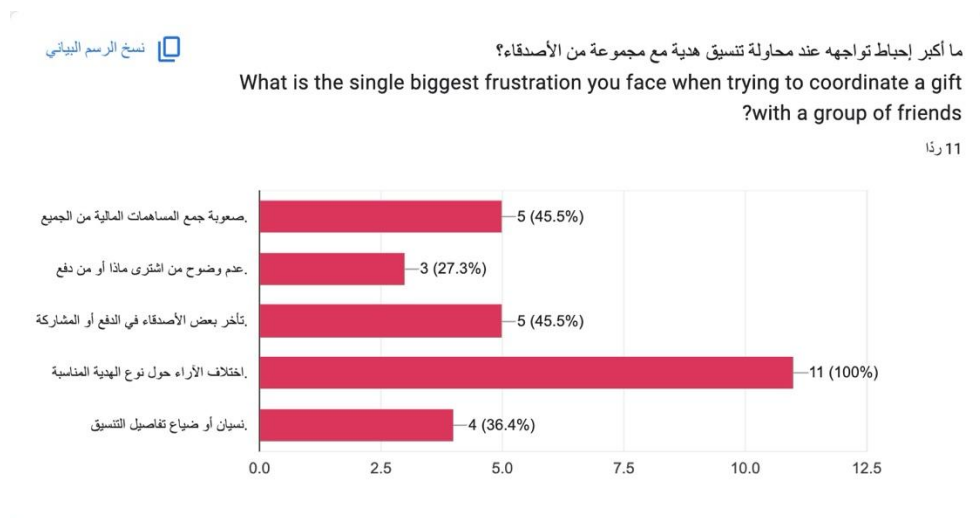


Figure 15: Fifth question in questionnaire.

Q6:

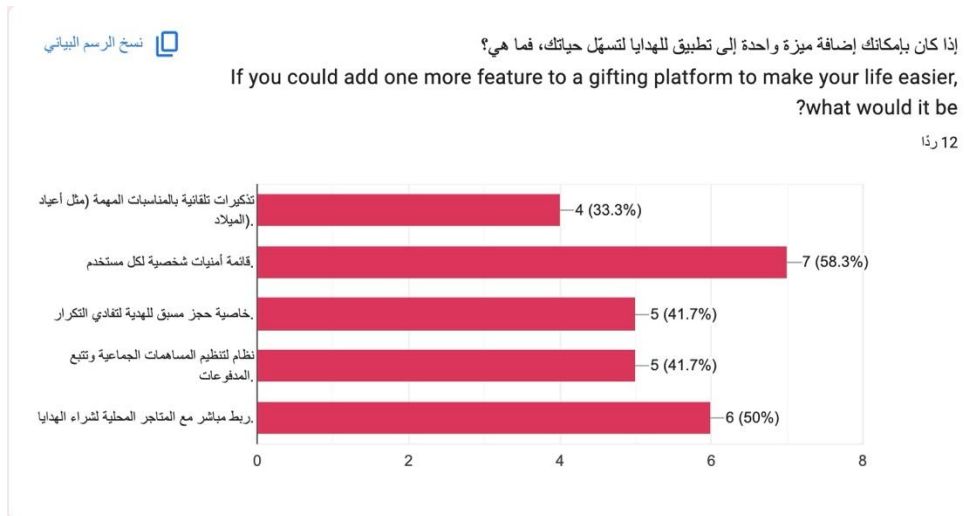


Figure 16: Sixth question in questionnaire.

8.2.2 Link to questionnaire responses:

https://docs.google.com/spreadsheets/d/1jHcsiJ87O1pE4wjmrja9u_C2HBrwfezqd9EagIDUaMY/edit?resourcekey=&gid=778336573#gid=778336573

8.3 Appendix_C: Jira

2025-IT320--Section79667-G1

Search bar... Filter

ID	Description	Priority	Status
IT320-1	As a non-registered user, I want to register using my username, email, and pa...	1000	4
IT320-2	As a registered user, I want to log in using my username or email and passwor...	1000	3
IT320-3	As a registered user, I want to log out of my account to prevent unauthorized ...	1000	2
IT320-4	As a user, I want to fully manage my personal Wishlist (create, add, view, edit, ...	1000	4
IT320-5	As a user, I want to control the visibility of my Wishlist so that I can decide wh...	1000	3
IT320-6	As a friend, I want to reserve a gift on a friend's list so that I can prevent gift d...	1000	4
IT320-7	As a user, I want to easily contribute to a group gift so that coordinating paym...	1000	5
IT320-8	As a user, I want to add occasion dates (e.g., birthdays, anniversaries).	1000	3
IT320-9	As a user, I want to search for friends by username so that I can view their pu...	1000	3
IT320-10	As a user, I want a direct button to add a post from the Explore feed to my pe...	1000	4
IT320-11	As a user, I want to Like an idea on the Explore feed.	1000	4
IT320-12	As a Store, I want to include a link to my external shop/contact page so that I...	1000	2
IT320-13	As a user or store, I want to post photos and descriptions of gift ideas so tha...	1000	4
IT320-14	As a user, I want the system to be available at least 95% of the time so that I...	1000	3
IT320-15	As a user, I want the system to meet performance standards so that I can int...	1000	3
IT320-16	As a user, I want the system to be highly usable so that all types of users ca...	1000	2

Figure 17: Product Backlog of Medad website in Jira

8.4 Appendix_D: Participant Consent Form

User Acceptance Testing (UAT) - Participant Consent Form

Project Title: WOD: A Social Gifting Platform

Research Team: IT320 - Software Engineering Team, College of Computer and Information Sciences, King Saud University

Supervisor: Dr. Reem AlGifary

1. Introduction

You are invited to participate in a user acceptance testing session for the WOD web platform. WOD is a social gifting platform designed to help friends and family organize, coordinate, and personalize their gift-giving experience. This testing is part of our academic project for the IT320 Practical Software Engineering course.

2. Purpose of the Study

The purpose of this testing is to evaluate the usability, functionality, and overall user experience of the WOD platform from the perspective of our target users. Your feedback will help us identify areas for improvement before the system's final release.

3. What Will Happen During the Test

If you agree to participate, you will be asked to:

- Use the WOD website to complete a series of realistic tasks (e.g., creating an account, managing a wishlist, reserving a gift).
- Share your thoughts and feedback on your experience.
- Complete a post-test questionnaire.

The entire session is expected to last approximately 30-40 minutes.

4. Potential Risks and Discomforts

We do not anticipate any risks beyond those encountered in normal computer use. You may find some tasks challenging, but this feedback is valuable for us. You can take a break at any time.

5. Benefits

While there is no direct benefit to you, your participation will contribute to the development of a software product that aims to solve real-world problems in social gift-giving. You will also gain insight into the software testing process.

6. Confidentiality

Your personal information and individual responses will be kept strictly confidential. The data will be used only for the purpose of this academic project and will be presented in an aggregated, anonymous format in our final report (e.g., "60% of participants...").

7. Voluntary Participation

Your participation is entirely voluntary. You are free to withdraw from the testing session at any time, for any reason, without any penalty or negative consequences.

8. Consent

By signing below, you confirm that:

- You have read and understood the information above.
- You voluntarily agree to participate in this user acceptance test.

- You consent to the collection and anonymous use of your demographic data and feedback for the purposes outlined in this form.

Participant Information & Agreement (Please fill in the following information)

Full Name:*

Demographic Information (For analysis purposes only):

Age:*

18-20
21 - 35
36-50
51+

Gender:*

Male
Female

Technical Proficiency:*

Basic (Comfortable using apps and websites for everyday tasks)
Advanced (Experience with programming, design, or complex software)

Have you used a dedicated gifting or wishlist platform before (e.g., Amazon Wishlist, etc.)?*

Yes
No

Link:

[User Acceptance Testing \(UAT\) - Participant Consent Form](#)

8.5 Appendix_E: User Acceptance Testing Questionnaire

User Acceptance Testing Questionnaire for WOD

Thank you for participating in testing the WOD platform. Please complete the following tasks as guided. Afterward, answer this questionnaire based on your experience. Your feedback is crucial for improving the system.

Instructions: For each statement, please select the option that best reflects your agreement.

(Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

1. I found the process of registering a new account and logging in to be straightforward.

*

2. It was easy to navigate through the different sections of the website (e.g., from the Home feed to my Profile or Calendar).

*

3. The process of creating a post and adding it to my wishlist was simple and understandable.

*

4. The gift reservation feature is clear and I believe it would effectively prevent duplicate gifts.

*

5. The group gifting feature would make it easier to coordinate and collect money for a shared gift with friends.

*

6. I could easily find the information I was looking for (e.g., a friend's profile, a specific gift item).

*

7. The website's interface (buttons, menus, labels) was clear and easy to understand.

*

8. Using the WOD platform for gift planning made me feel confident and in control of the process.

*

9. The website's response speed was fast and I experienced no significant delays.

*

10. Overall, I am satisfied with my experience using the WOD platform and would recommend it to others.

*

Open-Ended Feedback (Optional but highly appreciated):

What did you like most about the WOD platform?

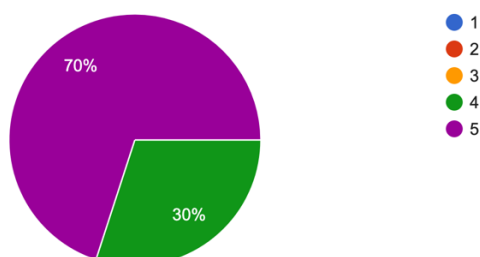
What features did you find confusing or difficult to use?

Do you have any suggestions for improvement?

8.5.1 User Acceptance Testing Questionnaire Results

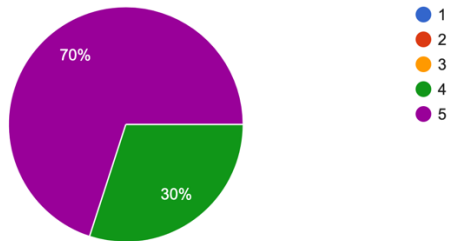
1. I found the process of registering a new account and logging in to be straightforward.

10 responses



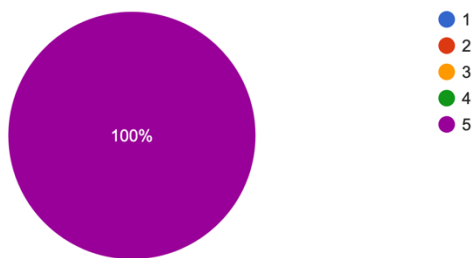
2. It was easy to navigate through the different sections of the website (e.g., from the Home feed to my Profile or Calendar).

10 responses



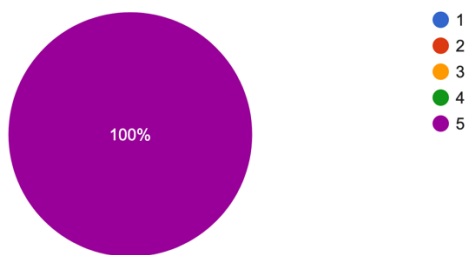
3. The process of creating a post and adding it to my wishlist was simple and understandable.

10 responses



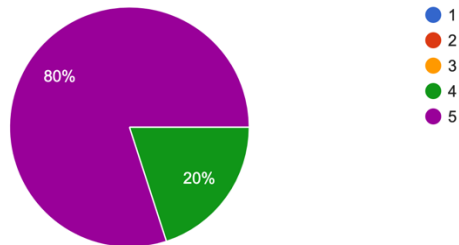
4. The gift reservation feature is clear and I believe it would effectively prevent duplicate gifts.

10 responses



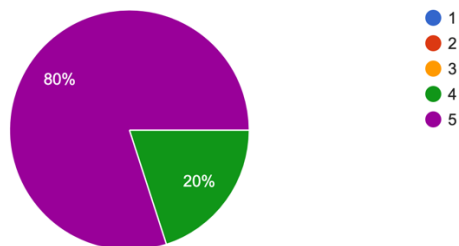
5. The group gifting feature would make it easier to coordinate and collect money for a shared gift with friends.

10 responses



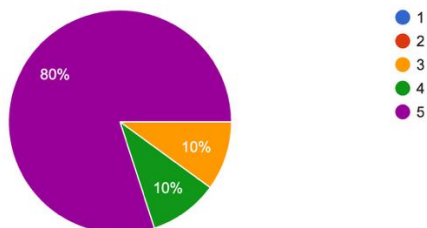
6. I could easily find the information I was looking for (e.g., a friend's profile, a specific gift item).

10 responses



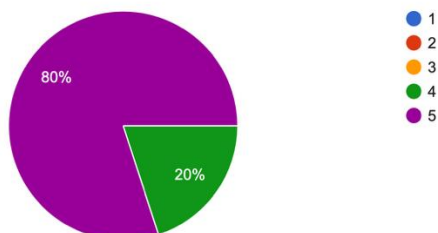
7. The website's interface (buttons, menus, labels) was clear and easy to understand.

10 responses



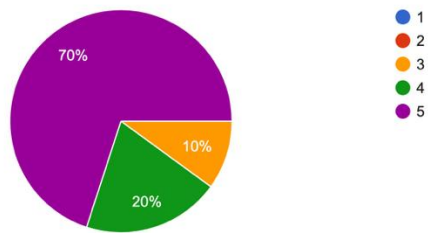
8. Using the WOD platform for gift planning made me feel confident and in control of the process.

10 responses



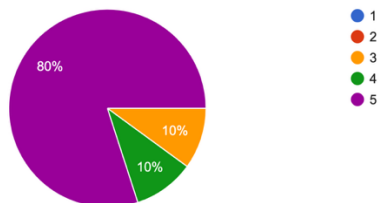
9. The website's response speed was fast and I experienced no significant delays.

10 responses



10. Overall, I am satisfied with my experience using the WOD platform and would recommend it to others.

10 responses



Link : [User Acceptance Testing Questionnaire for WOD](#)